



duolingo

The Art of Learning, Modernized

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LONG: Duolingo (NASDAQ: DUOL)

'25 Price Target: \$176.13 (85% upside, 21% IRR)

Business Overview

“A language learning disruptor poised for rapid monetization”

Duolingo (NASDAQ: DUOL) is a “freemium” gamified language-learning mobile app with over 500 million downloads globally.

The company generates revenue across 4 reporting segments...

1 Subscription (72% revenue)

- DUOL generates revenue from annual and monthly subscriptions to their premium service – Duolingo Plus
- We believe subscription penetration growth is understated due to a misunderstanding of DUOL’s core customer cohort, and is in fact set to hit a major inflection point

2 Advertising (15% of revenue)

- DUOL generates revenue from in-app advertising targeted at its free users
- We believe an ever-improving UX fueled by a powerful data moat will drive rapid growth in ad\$/user

3 English Testing (10% of revenue)

- DUOL generates revenue from the Duolingo English Test, an online on-demand assessment of English Proficiency accepted by 18 out of the T20 undergraduate programs in the US
- We found DUOL has seized 10% market share from incumbents within 2Y, indicating a superior product

4 Other (3% of revenue)

- DUOL generates revenue through in-app purchases of gems, streak repairs and other bonuses
- We believe in-app \$/DAU has huge runway for growth based on comps with other mobile apps

Key Financials

Cap Table	
Current Share Price	\$95.23
(x) FDSO	40
Market Cap	\$3,775
(+) Debt	30
(-) Cash	591
Enterprise Value	\$3,213
Shares Sold Short	3.3
% short float	8.27%
Total Shares Sold Publically	39
% float	97%

Summary of Key Stats & KPIs				
Years	2019A	2020A	2021A	2022E
Revenue	71	162	251	368
% y/y	194%	129%	55%	47%
Gross Profit	50	116	182	265
% margin	71%	72%	72%	72%
% y/y		131%	57%	46%
Adj. EBIT	(8)	3	(4)	5
% margin	(11%)	2%	(2%)	1%
% y/y		(143%)	(217%)	(220%)
Unlevered FCF	(1)	16	21	17
% margin	(1%)	10%	8%	5%
% y/y		(1675%)	25%	(19%)

Best-In-Category Product

“Formidable data moat powers rapid user growth”

Management’s use of **A/B testing** leverages customer base to constantly **improve UX** and **drive further monetization**

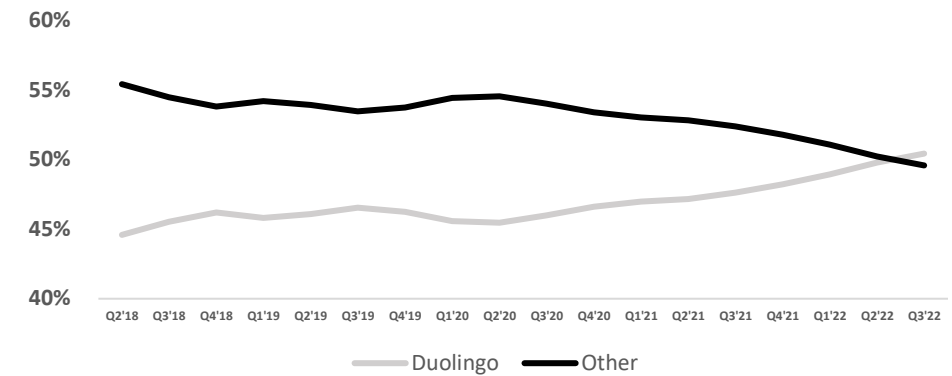
- DUOL **runs more than 500 A/B tests a quarter**, constantly finding ways to change the UX to increase monetization
- Improved UX drives more customer engagement and user growth, which in turn further **fuels DUOL’s powerful data moat** and allows it to **spread R&D costs across a larger revenue base**
- The result? **DUOL has gained more than 6% market share since 2018**, making it the **#1 education app in the world**

We believe DUOL benefits from a **high-quality management team**, helping us **build conviction in successful long-term execution**

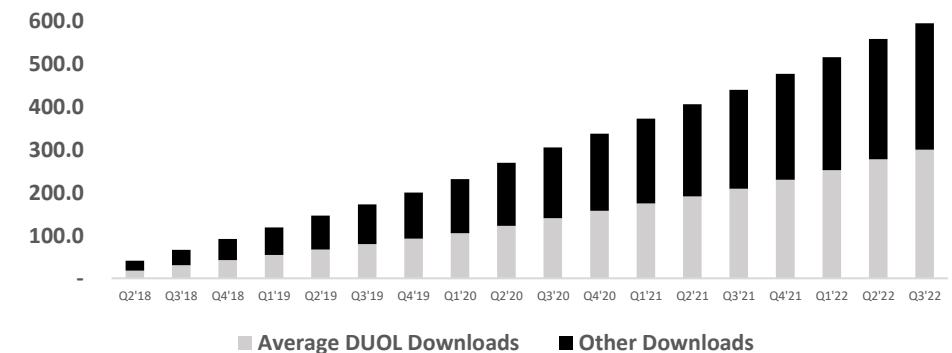
- **Innovative mgt team & data-driven CEO** (former reCAPTCHA CEO) combined with elite strategy team of almost all ex-MBB consultants
- **Product-first, “customer over everything mentality”** drives **continuous UX improvements** and engagement/monetization growth
- **Amazing company culture** leads to **top talent acquisition/retention**

High-Quality App Gaining Incremental Market Share

% App Download Market Share



Cumulative Language App Downloads



Investment Thesis

“Street mismodels long-term growth story despite expanding moat”

1

Misunderstanding of Core Customer Base

Subscription penetration should accelerate rather than decelerate post-COVID

2

Underestimated Subscription Penetration Runway

Under-monetized COVID Cohort masked “true” penetration growth

3

International Monetization is the Final Growth Lever

Flexible pricing plans are set to capitalize on un-tapped monetization base



1a) Misunderstood Core Customer

Street is pricing in **tapering down of subscription penetration growth post-COVID** based on a supposed “normalization” from re-opening

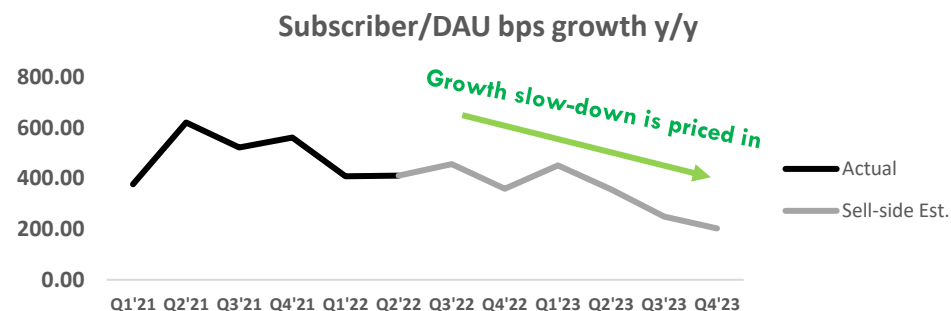
- “The **verdict is still out** on what a **normalized MAU growth rate** looks like for Duolingo **post the COVID surge**.” – Evercore 3/3/22
- “**Key risks to our Neutral rating include...**c) **Impact of COVID-19** (re-opening vs. additional stay-at-home orders) **in terms of engagement trends**” – GS 5/8/2022

The **idea that COVID was a tailwind for DUOL is a misconception...**in reality **COVID was a headwind for key monetization metrics.**

- DUOL users can be split into two main types: **motivated and casual**. These roughly correlate to users learning English, and users learning non-English.
- **Type 1 User**: “They’re primarily people learning a second language, **more for leisure, more for fun, more for travel** because it’s **not as much of a necessity**.”
- **Type 2 User**: “But a bit over half of the other users are learning English as a **form mainly of social mobility**, as a **way to get a better life...**English is a **necessity for them to have a career** and so on.” – Former DUOL Chief of Staff
- Our Thesis: **During COVID, DUOL’s user mix shifted from Type 2 motivated users to Type 2 casual users, creating a headwind for monetization #s**

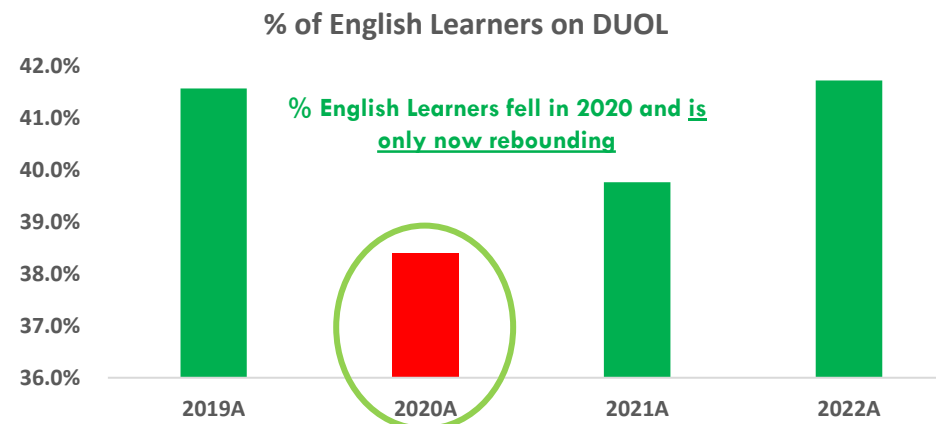
“DUOL is a COVID rebound, not a COVID darling”

Sell-side Estimates for Penetration Growth (VA)



Alternative Data Collection indicates temporary shift in customer mix during COVID

Using web-scraped data from DUOL, we found that the % of English learners (ie more “motivated”) declined during COVID



1b) Misunderstood Core Customer

“DUOL is a COVID rebound, not a COVID darling”

70 % of new learners who picked up a language during the pandemic started learning for family/culture reasons (as opposed to school/work)

Significant new user growth was driven by interest in Asian languages (Korean etc), which typically skew towards recreational learning motivations

- While the pandemic drove a increase in new downloads and active users, the mix of new users that joined the platform were predominantly “casual”

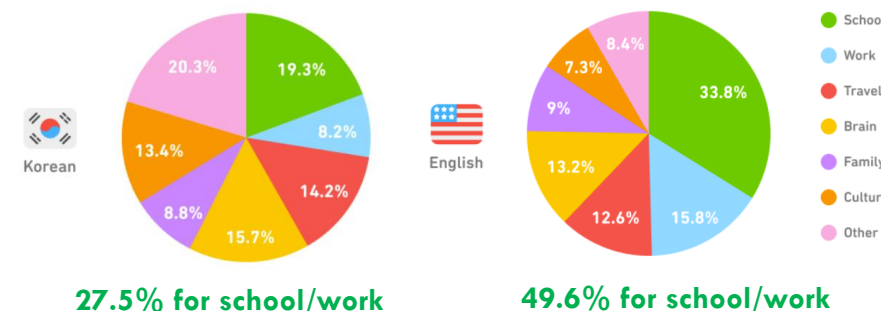
Only 23% of 2020 Incremental Users were English Learners...

	2020A	2021A	2022A
% English Share of New Learners	23.30%	52.30%	50.10%
% Non-English Share of New Learners	76.70%	47.70%	49.90%

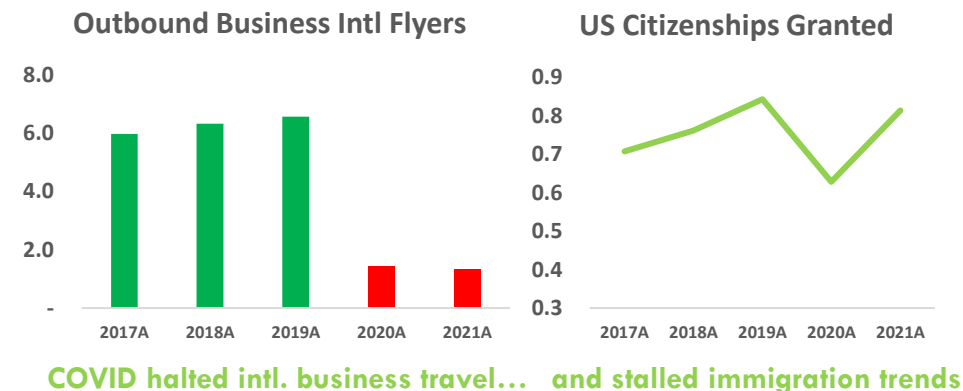
- This cohort was therefore less monetizable, had a higher degree of churn, and bought less in-app purchases/subscriptions
- Meanwhile, macro trackers associated with “motivated” learners declined
- Overall, COVID caused a drag on subscription penetration and other monetization KPIs

Conclusion: Although the Street believes monetization should decelerate post-COVID, we believe that it will in fact accelerate!

Comparing Motivations For English and Non-English Learning



Macro Trackers Show Motivated Learning SAM Declined During COVID

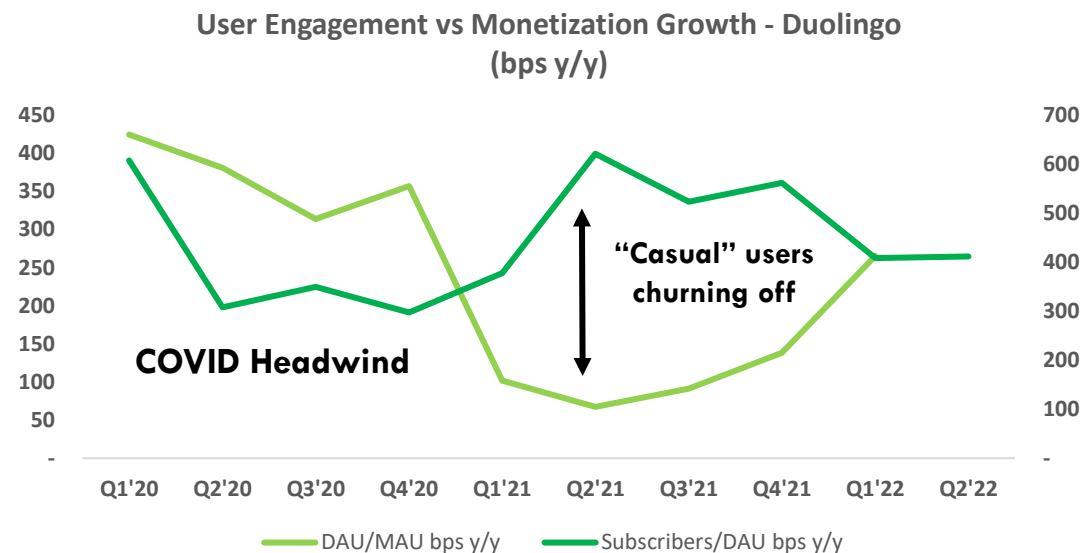
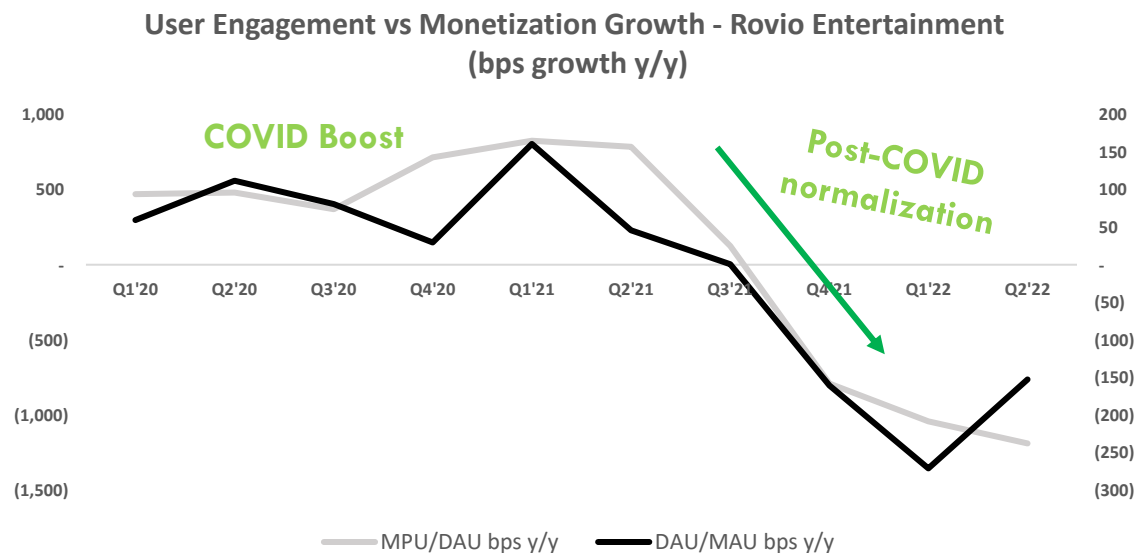


2a) Subscription Penetration Runway

“Hidden penetration growth is set to reach an inflection point”

Other Mobile Gaming comps saw monetization KPIs benefit from a COVID boost...DUOL is assumed to be the same

However, the opposite is true...an influx of casual learners obscured DUOL's “true” monetization growth



During 2020, most mobile games saw monetization trend upwards with user engagement and back downwards in 2021-22
(Rovio Entertainment: Angry Birds franchise)

In contrast, DUOL's DAU/MAU activity has declined since FY21, but monetization penetration accelerated despite tough comps

Conclusion: Unlike most mobile games, the cohort of new learners joining DUOL during COVID were less likely to pay than the original user base, which obscured “true” monetization growth

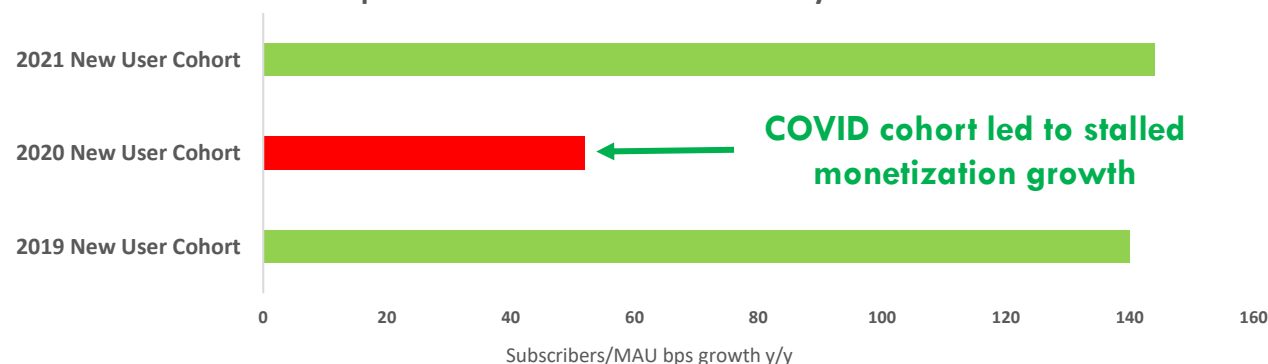
2b) Subscription Penetration Runway

“Hidden penetration growth is set to reach an inflection point”

To build conviction in our thesis we...

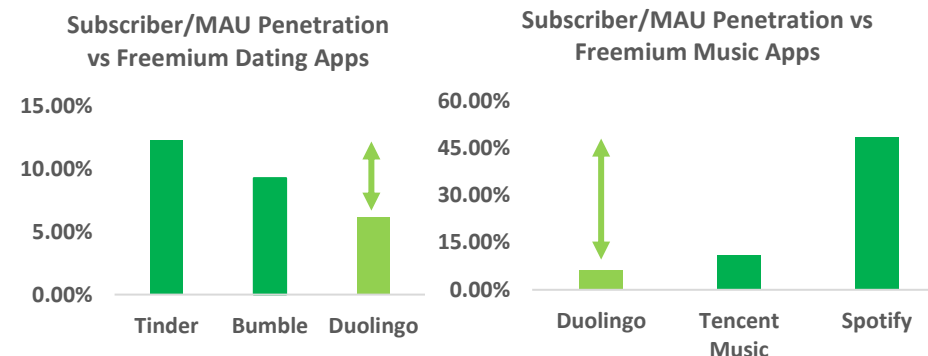
- 1 Backed into MAU and subscription churn using management quotes and benchmarking to industry averages
- 2 Factored in a decreased rate of MAU churn during COVID to reflect increased user engagement and time spent on app
- 3 Estimated subscription monetization by cohort (eg. learners who joined in 2020 vs 2019/21 etc.)

Comparison of Monetization Growth by User Cohort



Conclusion: Change in Customer Mix indeed led 2020 New User Cohort to be under monetized relative to historicals

In addition, freemium mobile app comps suggest long runway for subscription growth



Substantial room for growth compared to both categories

While the Street is focused on absolute #s, we are variant on the first derivative of growth

- Other investors look at **absolute growth numbers**, which are **up and to the right** (a testament to product & mgt quality)
- We observed COVID's effect on **stalling the rate of change of DUOL's growth trajectory**
- We are **variant on acceleration/deceleration**: DUOL is set to hit an upward inflection point instead of a downward one
- Ultimately, this is enough to put us **380bps above the street on subscriber penetration by 2026**

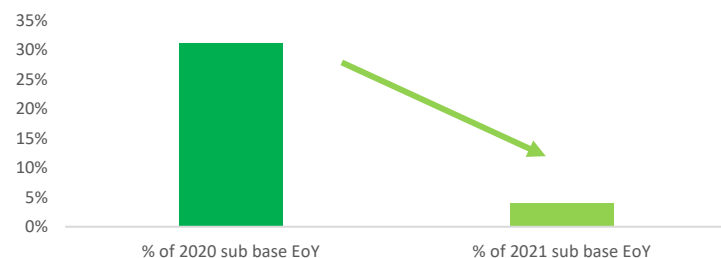
2c) Subscription Penetration Runway

“Hidden penetration growth is set to reach an inflection point”

COVID cohorts are now beginning to churn off the app and be replaced by “motivated” learners

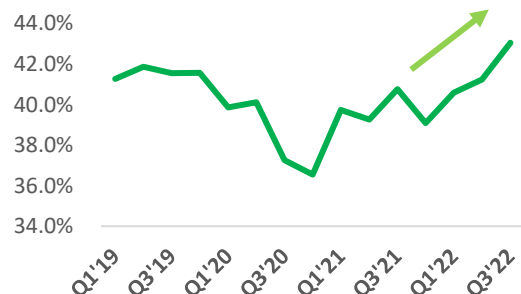
In past 4 quarters, “motivated” learning SAM has demonstrated dramatic recovery

COVID Cohort as a % of EoY subscriber base

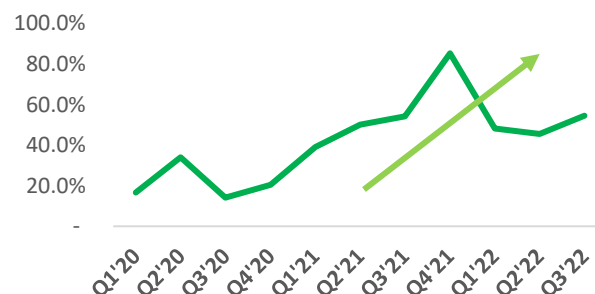


Unmotivated learners have been churning off the app...by EoY 2021, COVID cohort was only 4% of DUOL's total sub base.

% English Learners on DUOL

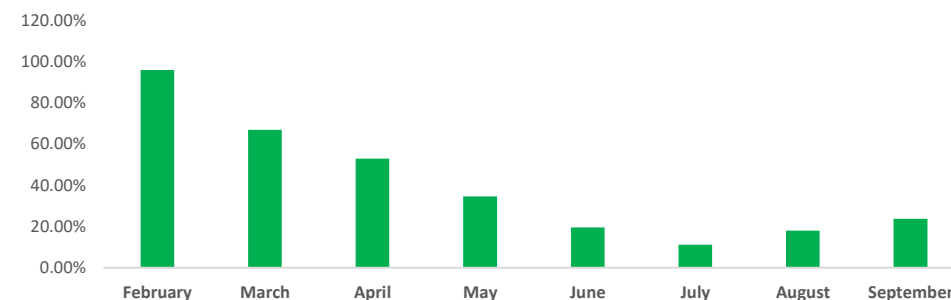


% English Learners of New Users



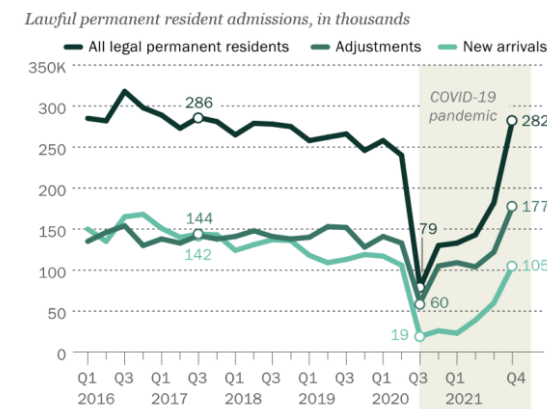
Overall & incremental customer mix has improved substantially in recent quarters...with COVID headwinds disappearing DUOL should be poised for re-accelerated growth

TSA Checkpoint Passengers YoY Growth



International travel experienced strong recovery particularly in 1H22...

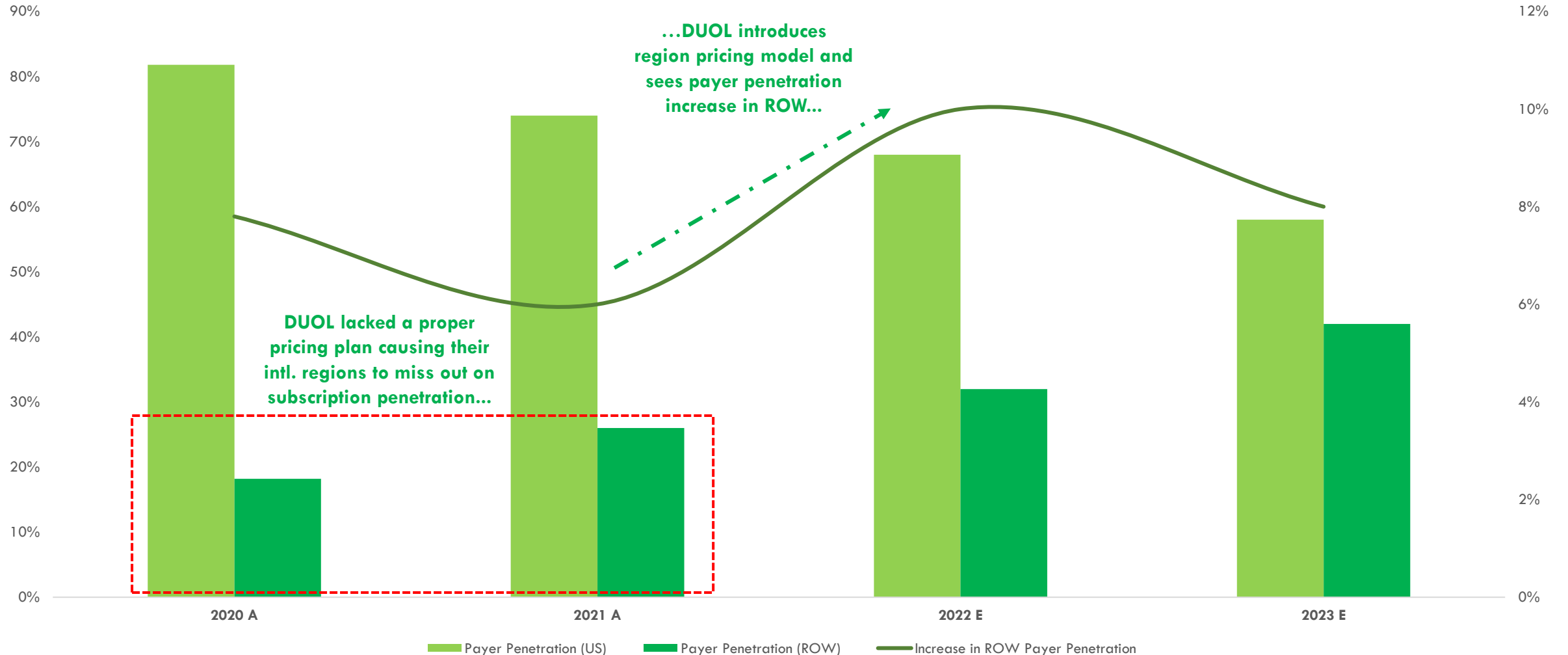
...and the US has seen a sharp rebound in immigration trends following the reopening of intl. borders



3a) International Monetization

“Two steps backwards, one leap forward”

ROW Is Underpenetrated in terms of Monetization



3b) International Monetization

“Two steps backwards, one leap forward”

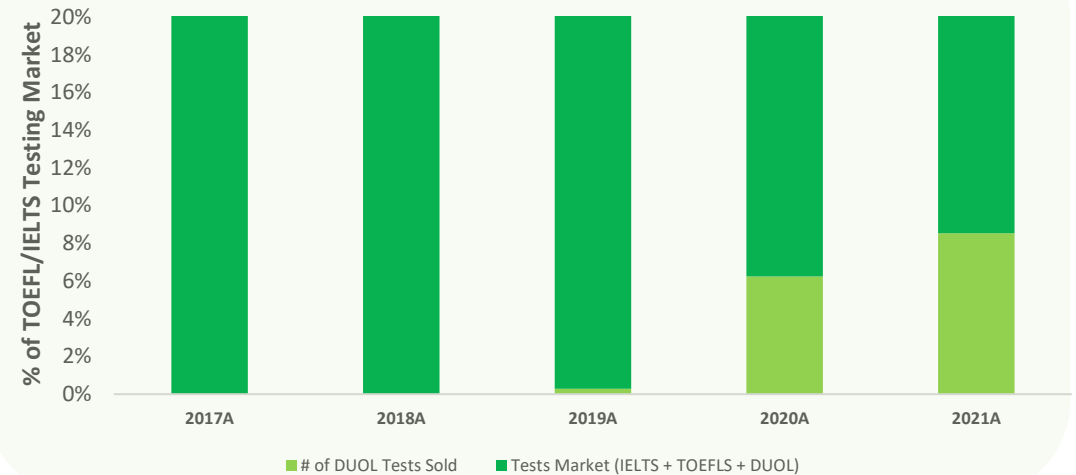
Flexible Pricing Plans Are Set to Drive RoW Monetization

- DUOL **IPO'd with no international pricing plan**. International users were under monetized and churned off the platform quickly...
- **...DUOL is rolling out flexible pricing plans respective to each country**
- “So the majority of countries, we've been testing reducing prices, and that's been happening. And it is having an impact “ Q1 22 Earnings Call
- Public comps (NFLX, SPOT, etc). serve as a basis for understanding the ability to monetize other major regions...

DUOL Testing Segment increases appeal to highly monetizable “motivated” learners

- DUOL has positioned itself to be more than a language learning app
- Abroad they give access to show **motivated learners** demonstrated ability in a language (typically English)
- Testing arm begins to take share away from standard tests as users' study from DUOL + use DUOL certification (accepted by X amt of US schools).

DUOL Testing Market Share Growth



DUOL has an **underpenetrated & underappreciated** opportunity to monetize **motivated users** in the ROW region

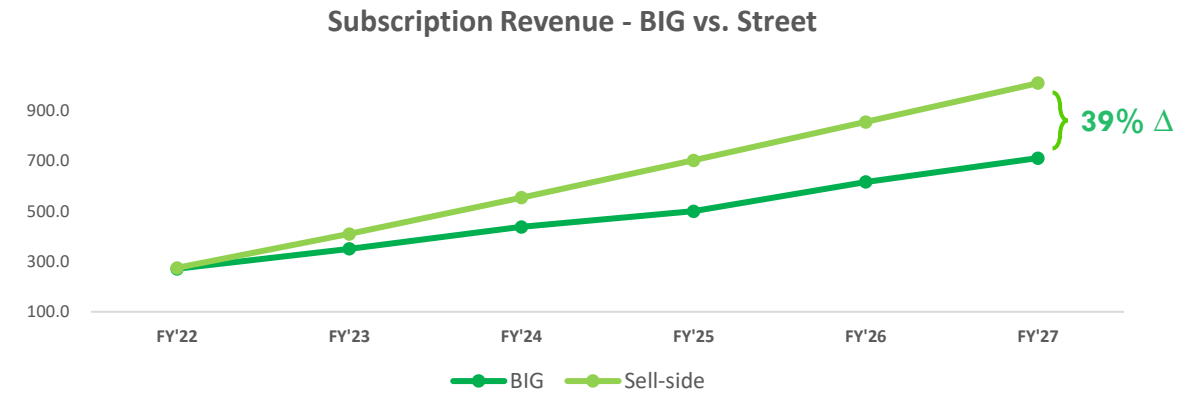
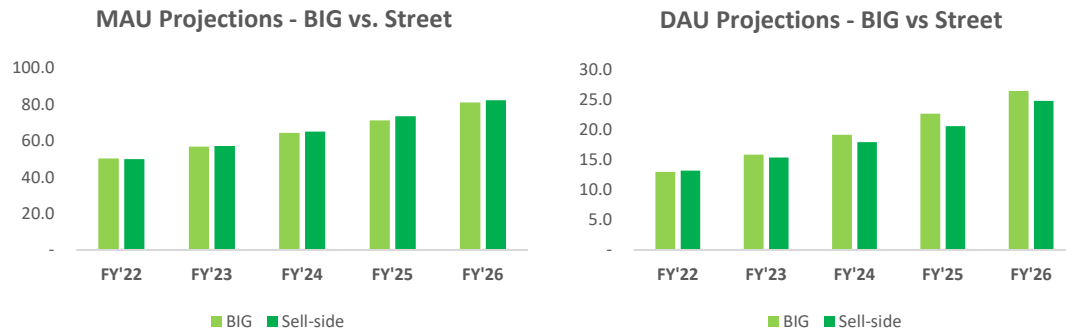
Variance to Street Analysis

“While the Street is looking at absolute #s, we observe the first derivative”

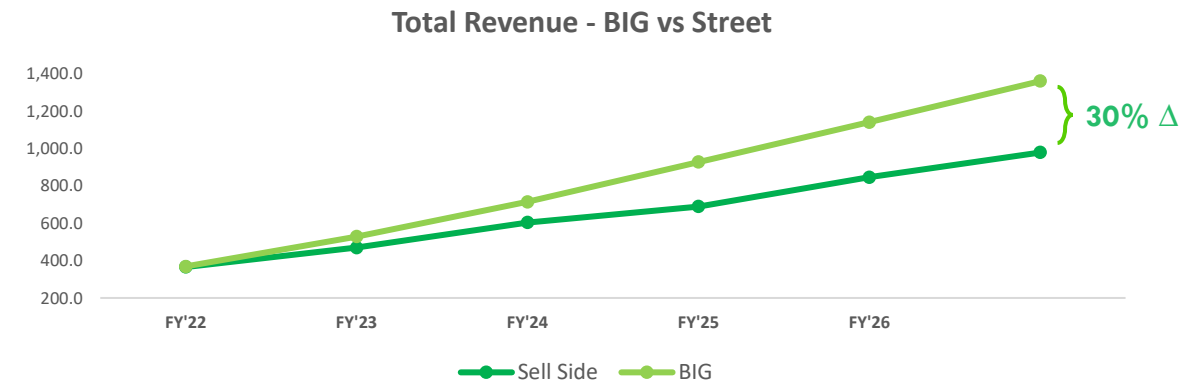
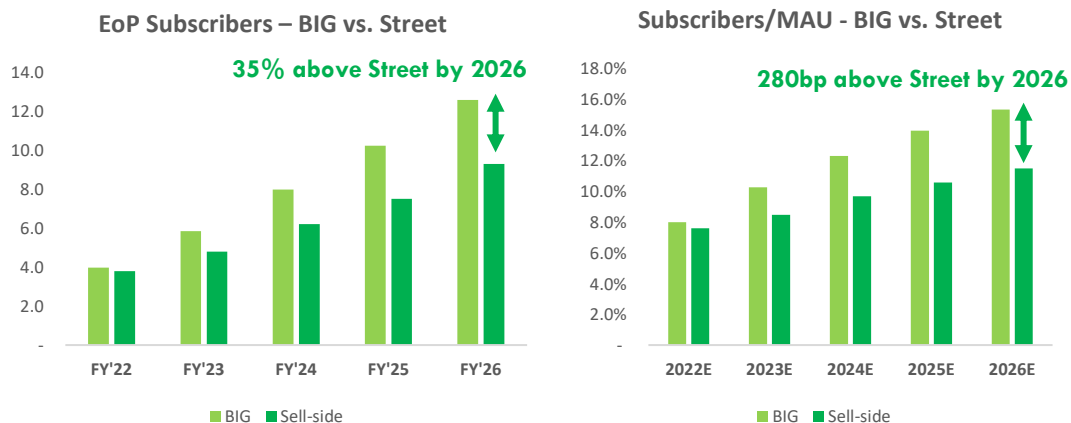
In summary, our thesis is that DUOL's monetization potential is heavily overlooked by the Street

Our variance on KPI projections flows into significant top-line beats in the next 5 years which can materially move the stock

Consequently, while we are in line with Sell-side on user growth...



We are variant on subscription penetration and subscriber growth



We are 39% above Street on Subscription Rev by 2026; 30% above on Total Rev

DCF & Year 5 Rev Exit Mult.

“While the Street is looking at absolute #s, we observe the first derivative”

Valuation Method #1: DCF Analysis (Base Case)

Fiscal Date	Historical			Estimates																		
	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E
Revenue	71	162	251	368	527	714	926	1,139	1,359	1,581	1,790	1,992	2,186	2,366	2,526	2,659	2,759	2,821	2,877	2,935	2,993	3,053
Gross Profit	50	116	182	269	389	528	684	842	1,004	1,168	1,323	1,472	1,616	1,750	1,868	1,967	2,041	2,087	2,129	2,172	2,215	2,259
% GM	-	-	-	73.2%	73.7%	74.0%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%
EBIT	(8)	3	(4)	8	33	91	165	241	325	414	500	587	679	773	865	953	1,032	1,100	1,122	1,145	1,167	1,191
% EBIT	-	-	-	2.3%	6.3%	12.8%	17.8%	21.2%	23.9%	26.2%	27.9%	29.5%	31.1%	32.7%	34.2%	35.8%	37.4%	39.0%	39.0%	39.0%	39.0%	39.0%
(-) Taxes	-	(0)	(0)	(0)	(7)	(19)	(35)	(51)	(69)	(87)	(105)	(124)	(140)	(159)	(177)	(194)	(209)	(221)	(224)	(228)	(231)	(234)
NOPAT	(8)	3	(4)	8	26	72	130	190	257	327	394	463	539	614	688	759	823	879	898	917	937	957
(+) Reinvestment	7	13	25	14	16	45	46	55	60	67	74	78	54	47	37	26	14	-	-	-	-	-
UFCF	(1)	16	21	22	42	116	176	245	317	393	468	541	593	661	726	785	837	879	898	917	937	957
Discount Rate				1.0	1.1	1.2	1.3	1.4	1.6	1.7	1.9	2.0	2.2	2.4	2.7	2.9	3.2	3.4	3.8	4.1	4.5	4.9
PV FCF				22	38	96	133	169	201	229	250	264	266	271	273	271	265	255	239	224	210	196

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Mature State (2035+)	Terminal Value Calculation
2040E+	Last Year UFCF
Incremental Decel	LT Growth Rate
(1.5%)	TV FCF
	TV
	PV of TV
% GM	Phase 1 PV FCF
74.0%	Implied EV
	(+) Cash
% GM & % EBIT Delta	(-) Debt
35.0%	Implied MC
	(/) FDSO
Eventual Tax Rate	Implied DCF PT
21.0%	% Δ

Valuation Method #2: Year 5 Exit Mult.

		Historical					Estimates (Annual)									
Calendar Year		#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	
Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	
Fiscal Date	Units	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	
Total Revenue		-	-	71	162	251	368	527	714	927	1,139	1,359	1,581	1,790	1,992	
% q/q																
% y/y					129%	55%	47%	43%	35%	30%	23%	19%	16%	13%	11%	
% 2y						254%	127%	110%	94%	76%	60%	47%	39%	32%	26%	
% 3y							420%	226%	185%	152%	116%	90%	71%	57%	47%	
Street							365	468	603	688	845	978				
% q/q																
% y/y								28%	29%	14%	23%	16%				
% 2y									65%	47%	40%	42%				
% 3y										89%	80%	62%				

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Exit Multiple	
'26 Revenue	1,139
'25 NTM Exit Multiple	6.0x
Implied EV	6,835
(+) 2025 Cash	1,334
(-) 2025 Debt	26
Implied MC	8,143
(/) FDSO	43.6
Implied Exit Multiple PT	\$186.6
% Δ	96%

Freemium Comps Analysis

To quantify our thesis, we analyzed how other freemium mobile apps trade relative to MAU payer penetration...

	MAU Payer Penetration	Enterprise Value (mm)	EV/Payer Penetration bps	Avg. Rev per Subscriber
Spotify	48.55%	\$14,844	3.07x	\$57.7
Bumble/Badoo	6.16%	\$4,010	6.51x	\$261.8
iQIYI	19.34%	\$4,401	2.28x	\$47.8
Rovio	3.70%	\$232	0.63x	\$40.38
SciPlay	3.70%	\$439	0.47x	\$95.3

Duolingo (\$100.3 ARPS) is above the 75th percentile of the comp set (\$95.3)

“While the Street is looking at absolute #s, we observe the first derivative”

Street Year 5 Est. Subscribers/MAU: 11.50%
BIG Year 5 Est. Subscribers/MAU: 15.32%

Valuation Method 1: 75th Percentile Multiple

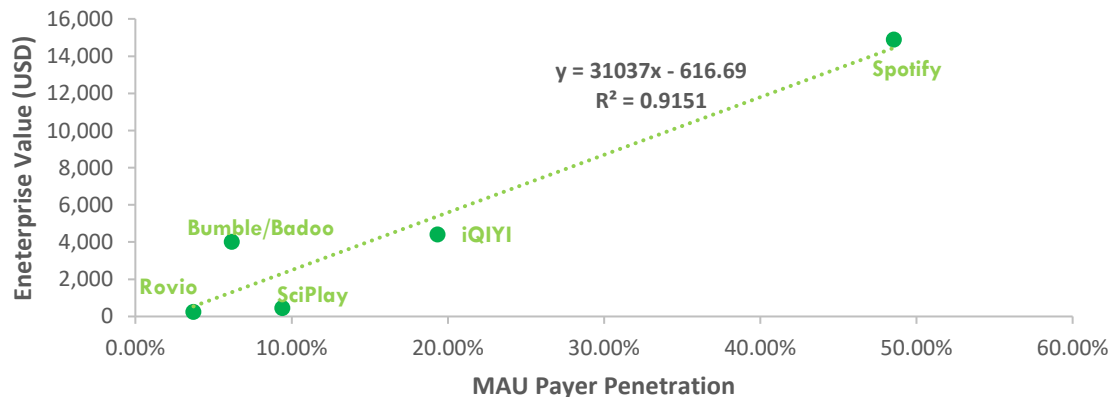
- Based on DUOL's ARPS, applying a 75th percentile multiple from our comp set of EV/Payer Penetration bps (3.07x)
- Multiplying by Street Est. penetration implies \$3.54bn enterprise value: **10.2% upside**
- Multiplying by BIG Est. penetration implies \$4.70bn enterprise value: **46.8% upside**

Valuation Method 2: Linear Regression

- Applying Street Est. penetration implies \$3.54bn enterprise value: **10.5% upside**
- Applying BIG Est. penetration implies \$4.67bn enterprise value: **45.8% upside**

Conclusion: Applying our projected penetration vs. Sell-side shows significant upside on a multiples basis

Freemium Comps - EV vs. MAU Payer Penetration



Blended Price Target

“While the Street is looking at absolute #s, we observe the first derivative”

'25 Valuation			
	Bear Case	Base Case	Bull Case
Exit Multiple '25 PT:	\$67.46	\$186.53	\$295.03
% irr	(10.1%)	23.0%	41.6%
% weight	7.5%	33%	7.5%
DCF PT:	\$72.61	\$199.83	\$319.16
% upside	(24%)	110%	235%
% weight	5%	20%	5%
Payor Penetration Multiple '25 PT:	\$97.49	\$137.59	\$143.21
% irr	0.7%	12.0%	13.4%
% weight	5%	7.5%	2.5%
Linear Regression '25 PT:	\$120.78	\$136.85	\$142.27
% irr	7.6%	11.8%	13.1%
% weight	2.5%	2.5%	2.5%
Blended PT:			\$176.1
% Δ:			85%
IRR:			21%

Pre-Mortem Analysis

“My investment went wrong. Why?”

Motivated Users Do Not Have a Higher Propensity to Spend

If motivated users do not display a higher propensity of spend (via. In-app purchases + subscriptions) our first thesis can be invalidated.

Mitigant: As explained by mgmt., there exists a large user base of motivated users who utilize DUOL for economic prosperity. With flexible pricing strategies we believe the value proposition that DUOL provides strongly incentivizes these consumers.

COVID Down-tic is Attributable to True Churn for Platform

If COVID down-tic is monetization is indicative of what true churn for DUOL is, it can hurt our projections for mature DUOL monetization.

Mitigant: There was an influx in casual users during the pandemic, leading to a lower-than-expected monetization value for DUOL. Our data indicates that these users have already churned off the platform and new users are more monetizable.

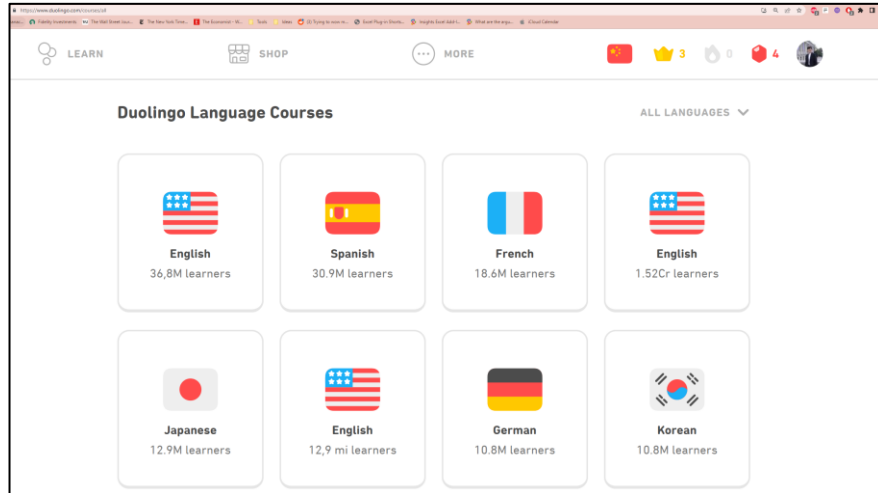
International Regions Not Composed of Motivated and Monetizable Users

If international regions are not comprised of motivated users, the monetization plans will not succeed slowing revenue growth for DUOL.

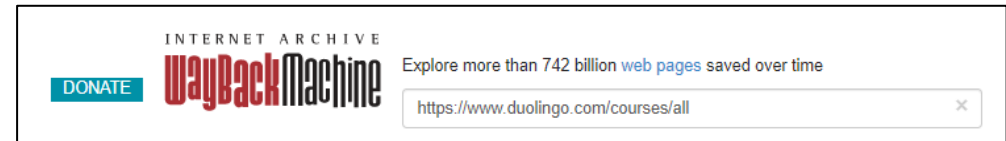
Mitigant: International users have shown to be highly-motivated (#1 language is English) and have demonstrated a demand for DUOL. The lack of monetization in ROW is a result of poor pricing which mgmt. is already combatting.

Thank You! Opening for Q&A...

Appendix 0: Data Collection, % English



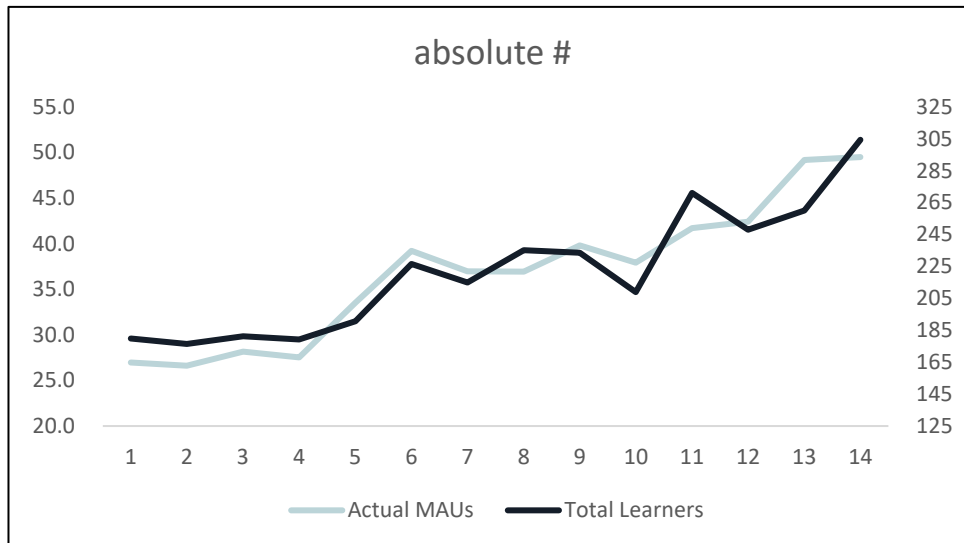
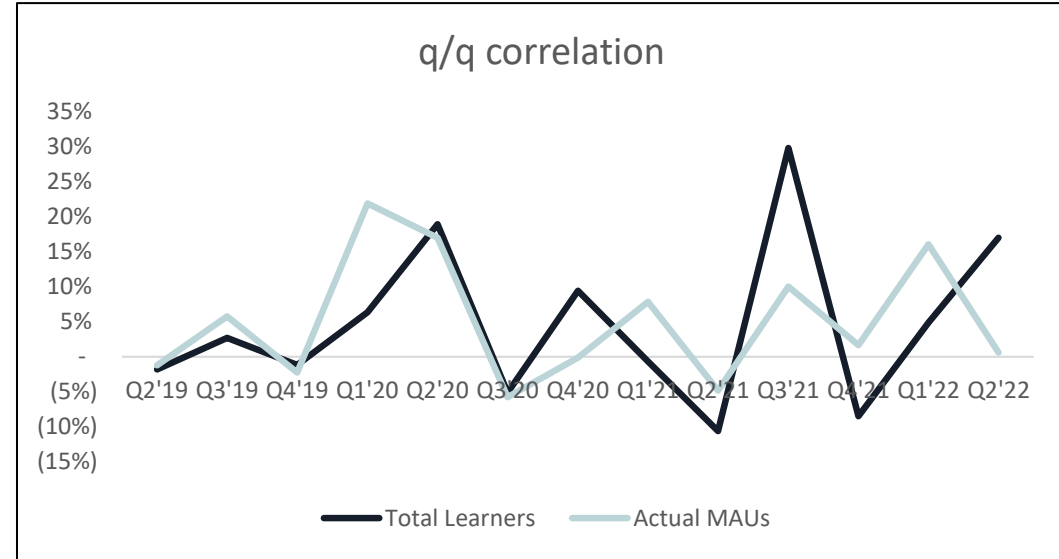
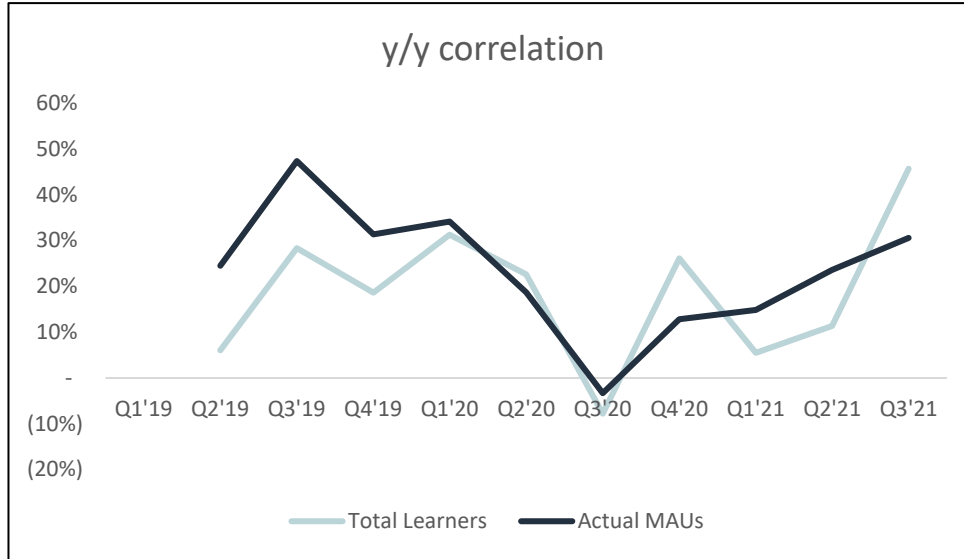
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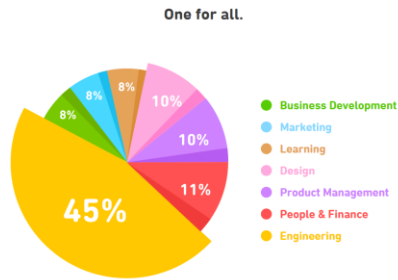
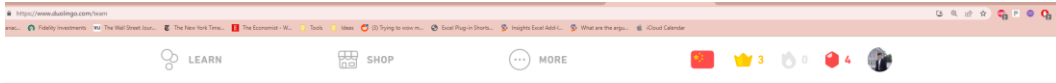
	Historical				Quarters															
	12/31/19	12/31/20	12/31/21	12/31/22	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	9/30/22	
	2019A	2020A	2021A	2022A	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	
Cumulative																				
English Learners	74	83	96	124	74	74	75	74	76	91	80	86	93	82	111	97	106	126	140	
% q/q						(0%)	2%	(1%)	2%	20%	(12%)	7%	8%	(12%)	35%	(12%)	9%	19%	12%	
% y/y		12%	15%	29%					2%	23%	6%	15%	22%	(10%)	38%	13%	14%	53%	27%	
Total Learners	179	217	240	297	180	177	181	179	191	227	215	235	234	209	271	248	260	304	325	
% q/q						(2%)	3%	(1%)	6%	19%	(5%)	9%	(1%)	(11%)	30%	(8%)	5%	17%	7%	
% y/y		21%	11%	23%					6%	28%	19%	31%	23%	(8%)	26%	5%	11%	46%	20%	
% English Share	41.6%	38.4%	39.8%	41.7%	41.3%	41.9%	41.6%	41.6%	39.9%	40.1%	37.3%	36.6%	39.7%	39.3%	41%	39.1%	40.6%	41.2%	43.0%	
bps q/q						59	(31)	1	(169)	24	(286)	(70)	318	(46)	149	(168)	151	64	181	
bps y/y		(318)	137	196				29	(199)	(144)	(431)	(332)	(38)	201	421	(65)	133	47	396	
Actual MAUs	23	27	37	40	26.9	26.6	28.2	27.5	33.5	39.2	37.0	36.9	39.8	37.9	41.7	42.4	49.2	49.5	50.0	
% q/q						(1%)	6%	(2%)	22%	17%	(6%)	(0%)	8%	(5%)	10%	2%	16%	1%	1%	
% y/y		17%	34%	10%					24%	47%	31%	34%	19%	(3%)	13%	15%	24%	31%	20%	
Learner to MAU Ratio	7.7x	7.9x	6.6x	7.3x	6.7x	6.6x	6.4x	6.5x	5.7x	5.8x	5.8x	6.4x	5.9x	5.5x	6.5x	5.9x	5.3x	6.1x	6.5x	
Incremental																				
Incremental English Learners		9	12	28					2	17	5	12	17	(9)	30	11	13	43	29	
% q/q										847%	(72%)	142%	46%	(153%)	(443%)	(64%)	16%	240%	(32%)	
% y/y			41%	128%								541%	(1%)	(286%)	164%	(35%)	(244%)	43%	168%	
Incremental Learners		38	24	56					11	50	34	56	43	(18)	56	13	27	95	54	
% q/q										363%	(33%)	66%	(23%)	(141%)	(417%)	(77%)	106%	260%	(43%)	
% y/y			(37%)	138%								418%	(14%)	(152%)	0%	(70%)	(250%)	70%	320%	
% Incremental English Share		23.3%	52.3%	50.1%					16.6%	34.0%	14.1%	20.5%	39.1%	50.1%	54.2%	85.4%	48.2%	45.5%	54.5%	
bps q/q										1,736	(1,982)	640	1,858	1,099	410	3,115	(3,713)	(270)	897	
bps y/y			2,907	(225)									2,252	1,616	4,008	6,483	912	(458)	29	

Appendix 0: % English Backchecked



Correlations		
0.50	% q/q	
0.66	% y/y	
0.94	absolute	

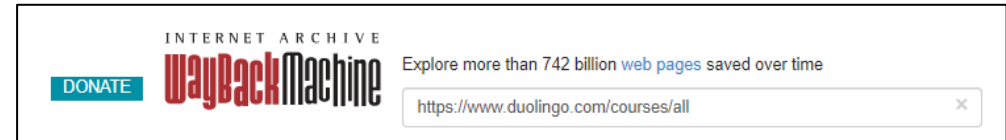
Appendix 0: Data Collection, Employees



We empathize with all our learners no matter how they are learning.

We offer podcasts, stories, and live events around the world across devices old and new.

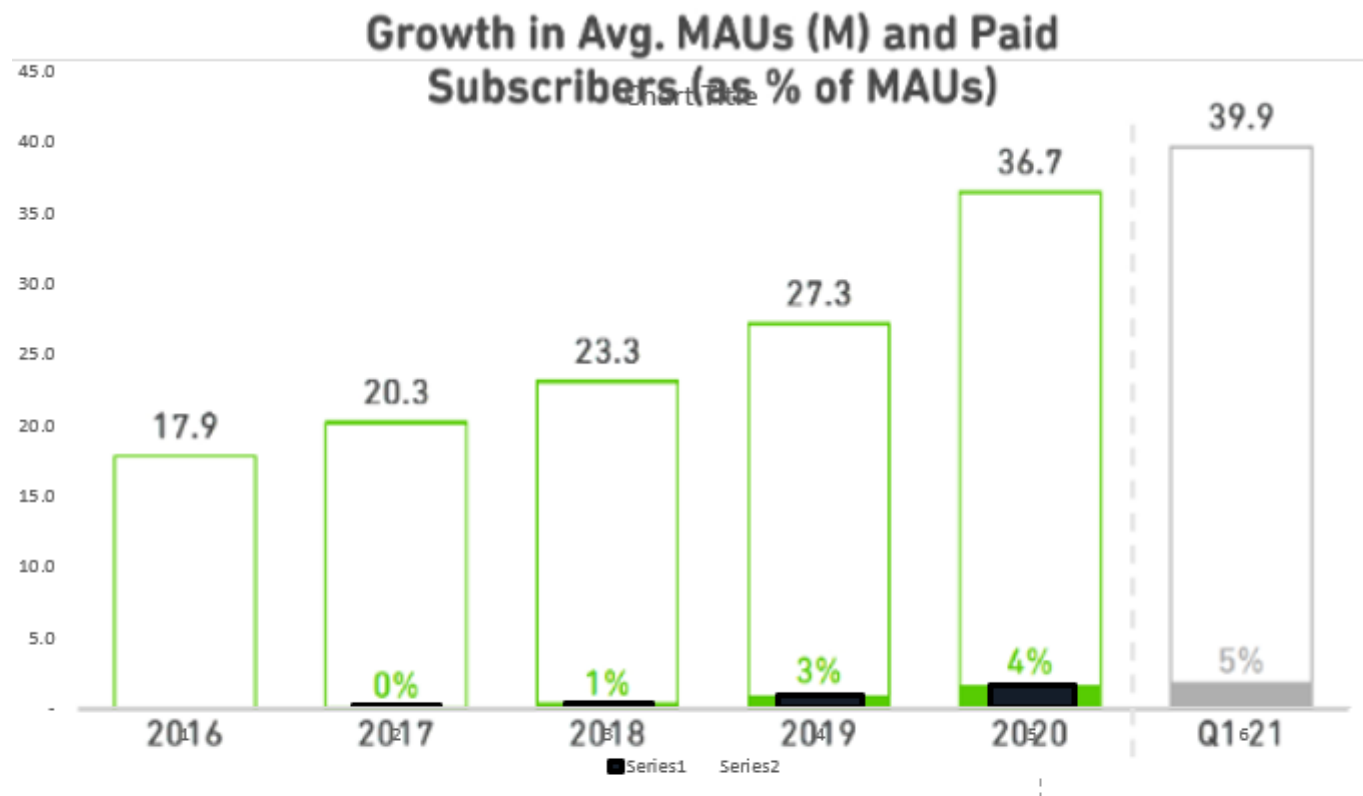
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Calendar Year	Quarters														
	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	9/30/22
Fiscal Date	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22
Implied Total Employees	200	220	220	230	240	250	333	360	381	425	464	490	544	613	613
% q/q		10.0%	-	4.5%	4.3%	4.2%	33.3%	8.0%	5.9%	11.4%	9.1%	5.7%	11.1%	12.5%	-
% y/y					9%	14%	45%	50%	53%	28%	29%	28%	28%	32%	25%
Research Employees (Adjuste	16	22	22	23	24	25	40	45	45	51	51	49	49	49	49
Engineers	106	110	110	115	120	125	165	175	182	199	213	221	245	276	276
R&D Employees	122	132	132	138	144	150	205	220	227	250	264	270	294	325	325
S&M Employees	12	13	13	14	14	15	20	23	25	29	33	36	44	49	49
People & Leadership	6	9	9	9	10	10	16	21	26	33	40	47	57	67	67
Product Management	6	-	-	-	-	-	5	10	16	23	31	40	52	61	61
Design	34	33	33	35	36	38	44	44	46	47	51	53	54	61	61
Operations	16	33	33	35	36	38	43	42	42	43	44	44	44	49	49
G&A Employees	62	75	75	78	82	85	109	118	129	146	166	183	207	239	239
Total Employees	196	220	220	230	240	250	333	360	381	425	464	490	544	613	613

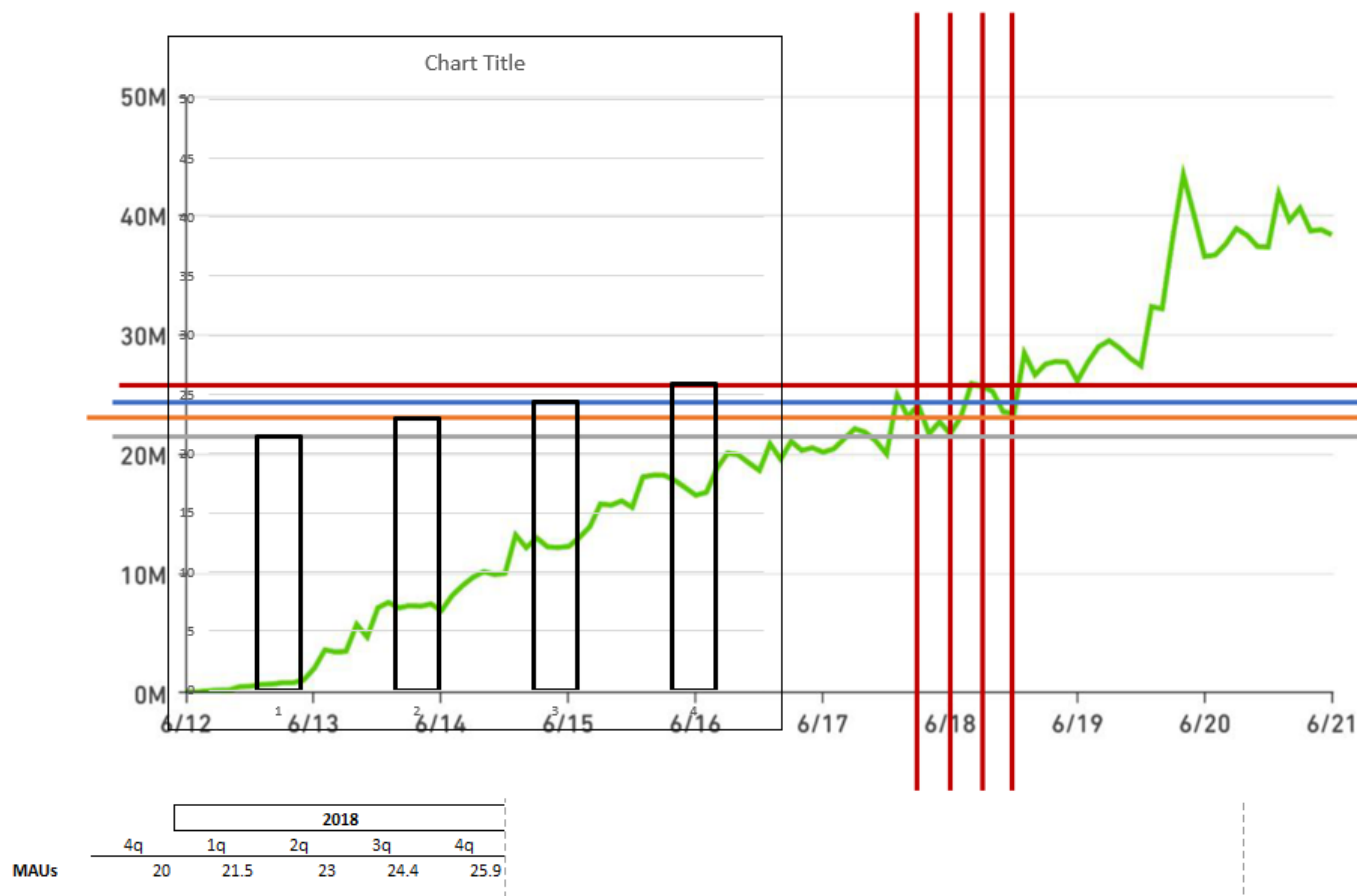
Appendix 0: Data Collection, <2017 Subs



	2016	2017	2018	2019	2020	2021
paid	-	0.1	0.4	0.9	1.6	-
% of total	0.00%	0.50%	1.50%	3.22%	4.41%	-
non-paid	17.9	20.2	23.0	26.4	35.1	39.9
% of total	100%	100%	99%	97%	96%	100%
total	17.9	20.3	23.4	27.3	36.7	39.9

Appendix 0: Data Collection, <2017 MAUs

Monthly Active Users (M)



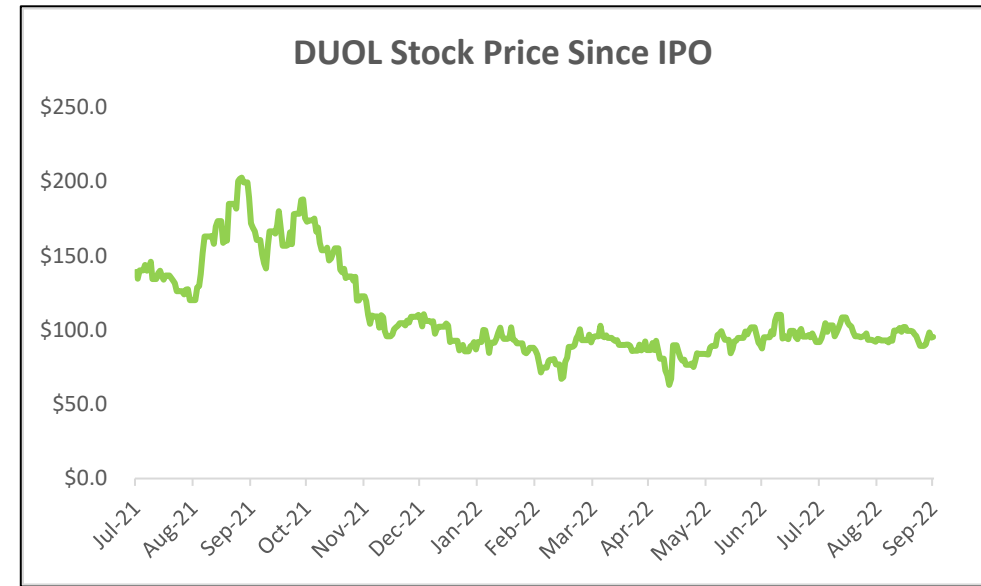
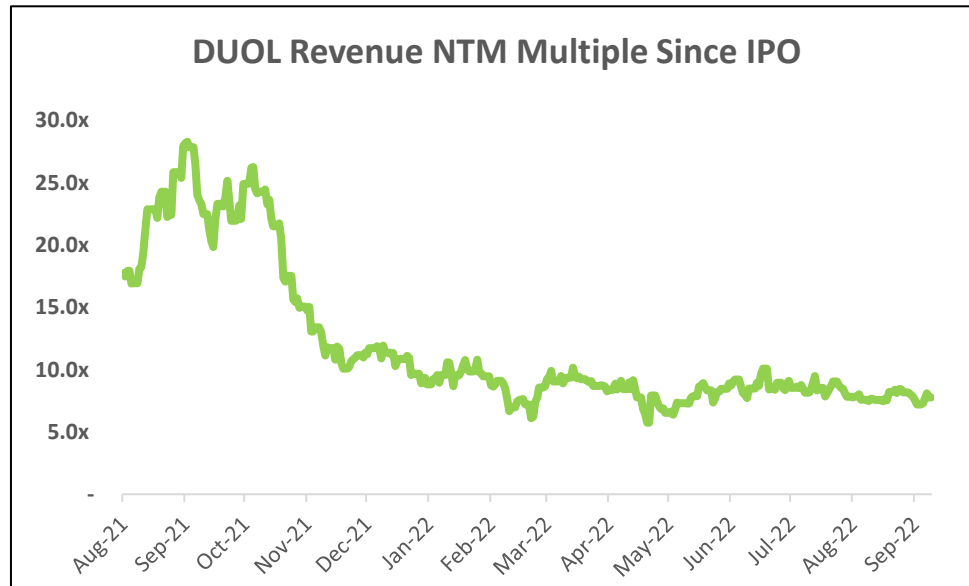
Appendix 0: Summary, Core Variance

		Consensus & BIG Differentiation														
Years	Units	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
BIG's Key Differentiations from the Street																
Subscription Revenue:																
Street	\$	72	75	80	85	91	93	270	348	433	492	602				
% y/y	%	56%	47%	37%	30%	25%	18%	8%	29%	24%	14%	22%				
BIG	\$	73	78	88	99	107	115	274	409	553	701	853	1,007	1,162	1,312	1,454
% y/y	%	58%	53%	51%	51%	48%	47%	9%	49%	35%	27%	22%				
Δ to Street %	%	1%	5%	10%	16%	18%	24%	2%	17%	28%	43%	42%				
% Paid Penetration (of MAUs)																
Street	\$	7.1%	7.5%	7.3%	7.8%	8.0%	8.3%	7.6%	8.5%	9.7%	10.6%	11.5%				
bps y/y	%	181	158	140	111	74	45	143	49	(59)	(173)	(246)				
BIG	\$	7.3%	7.8%	7.9%	8.9%	9.5%	10.1%	8.0%	10.3%	12.3%	14.0%	15.3%	16.5%	17.4%	18.2%	18.9%
bps y/y	%	201	194	204	224	225	226	181	228	203	165	136				
Δ to Street %	%	3%	5%	9%	15%	19%	22%	5%	21%	27%	32%	33%				
Other Street Estimates																
MAUs	\$	51	51	56	56	57	58	50	57	64	71	81				
% q/q	%	3%	0%	9%	(0%)	3%	1%									
% y/y	%	22%	20%	13%	12%	13%	13%	24%	13%	13%	11%	14%				
Total Revenue	\$	95	100	108	113	120	124	365	467	598	674	815				
% q/q	%	8%	6%	7%	5%	6%	3%	-	-	-	-	-				
% y/y		49%	38%	33%	28%	27%	24%	28%	28%	13%	21%	10%				
Gross Profit	\$	69	73	79	83	88	92	266	343	444	502	605				
% margin	%	72.4%	73.0%	73.5%	73.4%	73.3%	73.6%	73.0%	73.5%	74.2%	74.4%	74.3%				
Adj. EBIT	\$	(3)	1	6	6	3	5	7	21	53	97	171				
% margin	%	(2.8%)	0.5%	5.5%	5.3%	2.4%	4.4%	1.8%	4.6%	8.8%	14.4%	21.0%				
Diluted EPS Excl. Extra Items	\$	(0.06)	0.01	0.15	0.15	0.06	0.11	0.14	0.49	1.20	2.04	3.40				
% q/q	%															
% y/y	%															

Appendix 0: Summary, Key Financials

		Summary of Key Stats & KPIs														
Years	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Key Financials & Valuation (BIG)																
Subscription Revenue		2	10	24	71	162	251	368	527	714	926	1,139	1,359	1,581	1,790	1,992
% revenue		39%	90%	96%	78%	73%	72%	75%	78%	77%	76%	75%	74%	73%	73%	73%
% y/y			507%	141%	194%	129%	55%	47%	43%	35%	30%	23%	19%	16%	13%	11%
Advertising Revenue		-	-	-	15	28	39	45	50	71	98	131	168	209	243	279
% revenue		-	-	-	21%	17%	15%	12%	10%	10%	11%	12%	12%	13%	14%	14%
% y/y						85%	40%	16%	12%	42%	38%	33%	28%	24%	16%	15%
English Test Revenue		1	1	1	1	15	25	33	42	51	71	80	90	99	107	114
% revenue		61%	10%	4%	1%	9%	10%	9%	8%	7%	8%	7%	7%	6%	6%	6%
% y/y		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
In-App Revenue		-	-	-	-	2.0	6.9	15.9	26.2	39.3	55.8	74.5	93.7	111.5	128.5	144.8
% revenue		-	-	-	-	1%	3%	4%	5%	6%	6%	7%	7%	7%	7%	7%
% y/y							248%	130%	65%	50%	42%	34%	26%	19%	15%	13%
Revenue	\$	2	10	24	71	162	251	368	527	714	926	1,139	1,359	1,581	1,790	1,992
% y/y	%		507%	141%	194%	129%	55%	47%	43%	35%	30%	23%	19%	16%	13%	11%
Gross Profit	\$				50	116	182	269	389	528	684	842	1,004	1,168	1,323	1,472
% margin	%				71%	72%	72%	73%	74%	74%	74%	74%	74%	74%	74%	74%
% y/y	%					131%	57%	48%	44%	36%	30%	23%	19%	16%	13%	11%
Adj. EBIT	\$				(8)	3	(4)	8	33	91	165	241	325	414	500	587
% margin	%				(11%)	2%	(2%)	2%	6%	13%	18%	21%	24%	26%	28%	29%
% y/y	%					(143%)	(217%)	(307%)	293%	175%	81%	47%	35%	27%	21%	18%
Unlevered FCF	\$				(1)	16	21	22	42	116	176	245	317	393	468	541
% margin	%				(1%)	10%	8%	6%	8%	16%	19%	22%	23%	25%	26%	27%
% y/y	%					(1675%)	25%	7%	91%	176%	51%	39%	29%	24%	19%	15%
Implied EV/NTM Sales (BIG)	x							12.72x	9.40x	7.24x	5.89x	4.93x	4.24x	3.75x	3.37x	
Implied EV/NTM EBITDA (BIG)	x							202.46x	73.59x	40.76x	27.78x	20.62x	16.21x	13.43x	11.42x	
Implied EV/NTM UFCF (BIG)	x							159.12x	57.59x	38.13x	27.37x	21.16x	17.06x	14.32x	12.40x	
Implied EV/NTM Sales (Street)	x							6.88x	5.37x	4.77x	3.94x					
Implied EV/NTM EBIT (Street)	x							151.03x	61.19x	33.23x	18.80x					

Appendix 0: NTM Revenue & Price Hist.



Appendix 1: Scenario Analysis

Scenario Analysis

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Calendar Year	Estimates (Quarterly)						Estimates (Annual)								Steady State	
	9/30/22	12/30/22	3/30/23	6/30/23	9/30/23	12/30/23	12/30/22	12/30/23	12/30/24	12/30/25	12/30/26	12/30/27	12/30/28	12/30/29	12/30/30	
Year	2022	2022	2023	2023	2023	2023	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Year	2022	2022	2023	2023	2023	2023	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Quarter	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	
Fiscal Date	Units	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	FY'22	FY'23	FY'24	FY'25	FY'26	FY'27	FY'28	FY'29	FY'30	2030+

Chosen Scenario: **Base**

Growth:

MAU Growth	20%	20%	15%	14%	14%	14%
DAU/MAU Conversion	200	200	175	175	200	200

Monetization:

First-Year MAU penetration (bps y/y)	100	100	150	150	150	150
Second-Year MAU penetration (bps y/y)	50	50	140	140	140	140

Resulting Impact:

# of MAUs	50.0	50.9	56.6	56.4	57.0	58.0	50	57	65	73	82	91	100	109	118
% DAU/MAU	26%	26%	27%	28%	28%	28%	26%	28%	29%	31%	32%	33%	34%	35%	36%
bps y/y							202	188	160	145	125	110	100	90	80
% First-Year MAU Penetration	6.3%	6.2%	7.1%	6.9%	7.8%	7.7%	5.9%	7.4%	8.6%	9.7%	10.7%	11.5%	12.2%	12.7%	13.0%
% Second-Year MAU Penetration	7.3%	9.2%	9.1%	12.4%	8.7%	10.6%	8.7%	10.2%	11.4%	12.2%	12.7%	13.2%	13.5%	13.9%	14.1%
Paid Penetration (of DAUs)	28.6%	30.4%	29.2%	31.4%	34.7%	36.3%	30.9%	37.1%	42.0%	45.4%	47.8%	49.7%	51.1%	52.1%	52.8%
Paid Subscribers	3.3	3.6	4.0	4.5	5.0	5.4	4.0	5.9	8.0	10.3	12.6	15.0	17.5	20.0	22.4

Base Case

Growth:

MAU Growth	20%	20%	15%	14%	14%	14%
DAU/MAU Conversion	200	200	175	175	200	200

Monetization:

First-Year MAU penetration (bps y/y)	100	100	150	150	150	150
Second-Year MAU penetration (bps y/y)	50	50	140	140	140	140

Bear Case

Growth:

MAU Growth	17%	17%	12%	11%	11%	11%
DAU/MAU Conversion	175	175	150	150	175	175

Monetization:

First-Year MAU penetration (bps y/y)	50	50	100	100	100	100
Second-Year MAU penetration (bps y/y)	25	25	90	90	90	90

Bull Case

Growth:

MAU Growth	23%	23%	18%	17%	17%	17%
DAU/MAU Conversion	275	250	225	225	250	250

Monetization:

First-Year MAU penetration (bps y/y)	125	125	175	175	175	175
Second-Year MAU penetration (bps y/y)	75	75	165	165	165	165

14%	13%	12%	11%	10%	9%	8%
160	145	125	110	100	90	80

(1.5%) <- Incremental Growth (Post 2030)

74%
35%

<- Mature State GM

<- Mature State EBIT - GM Δ

12%	11%	10%	9%	8%	7%	6%
130	115	95	80	70	60	50

(2.0%) <- Incremental Growth (Post 2030)

74%
57%

<- Mature State GM

<- Mature State EBIT - GM Δ

16%	15%	14%	13%	12%	11%	10%
210	195	175	160	150	140	130

(1.5%) <- Incremental Growth (Post 2030)

74%
27.5%

<- Mature State GM

<- Mature State EBIT - GM Δ

today	9/30/22
comp pt	12/31/25

Valuation			
	Bear Case	Base Case	Bull Case

'26 Revenue (BIG)	939	1,139	1,279
(x) '25 NTM Multiple	2.0x	6.0x	9.0x
(+) Net Cash	1,032	1,308	1,455
(/) FDSO	43.1	43.7	43.9

Comps PT:	\$67.46	\$186.53	\$295.03
% irr	(10.0%)	23.1%	41.8%
% weight	15%	40%	10%
DCF PT:	\$72.61	\$199.83	\$319.16
% upside	(23%)	111%	236%
% weight	5%	25%	5%

Blended PT:	\$183.8
% Δ:	94%

Risk/Reward:

Appendix 1: Scenario Analysis

Scenario Analysis

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Calendar Year	Estimates (Quarterly)						Estimates (Annual)								Steady State	
	9/30/22	12/30/22	3/30/23	6/30/23	9/30/23	12/30/23	12/30/22	12/30/23	12/30/24	12/30/25	12/30/26	12/30/27	12/30/28	12/30/29	12/30/30	
Year	2022	2022	2023	2023	2023	2023	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Year	2022	2022	2023	2023	2023	2023	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Quarter	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	
Fiscal Date	Units	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	FY'22	FY'23	FY'24	FY'25	FY'26	FY'27	FY'28	FY'29	FY'30	2030+

Chosen Scenario: **Base**

Growth:

MAU Growth	20%	20%	15%	14%	14%	14%	14%	13%	12%	11%	10%	9%	8%	(1.5%)
DAU/MAU Conversion	200	200	175	175	200	200	160	145	125	110	100	90	80	74%

Monetization:

First-Year MAU penetration (bps y/y)	100	100	150	150	150	150	125	110	95	80	65	50	35	35%
Second-Year MAU penetration (bps y/y)	50	50	140	140	140	140	117	82	57	42	37	32	27	

Resulting Impact:

# of MAUs	50.0	50.9	56.6	56.4	57.0	58.0	50	57	65	73	82	91	100	109	118
% DAU/MAU bps y/y	26%	26%	27%	28%	28%	28%	26%	28%	29%	31%	32%	33%	34%	35%	36%
							202	188	160	145	125	110	100	90	80
% First-Year MAU Penetration	6.3%	6.2%	7.1%	6.9%	7.8%	7.7%	5.9%	7.4%	8.6%	9.7%	10.7%	11.5%	12.2%	12.7%	13.0%
% Second-Year MAU Penetration	7.3%	9.2%	9.1%	12.4%	8.7%	10.6%	8.7%	10.2%	11.4%	12.2%	12.7%	13.2%	13.5%	13.9%	14.1%
Paid Penetration (of DAUs)	28.6%	30.4%	29.2%	31.4%	34.7%	36.3%	30.9%	37.1%	42.0%	45.4%	47.8%	49.7%	51.1%	52.1%	52.8%
Paid Subscribers	3.3	3.6	4.0	4.5	5.0	5.4	4.0	5.9	8.0	10.3	12.6	15.0	17.5	20.0	22.4

today
comp pt

9/30/22
12/31/25

Valuation			
	Bear Case	Base Case	Bull Case

'26 Revenue (BIG)	939	1,139	1,279
(x) '25 NTM Multiple	2.0x	6.0x	9.0x
(+) Net Cash	1,032	1,308	1,455
(/) FDSO	43.1	43.7	43.9
Comps PT:	\$67.46	\$186.53	\$295.03
% irr	(10.0%)	23.1%	41.8%
% weight	15%	40%	10%
DCF PT:	\$72.61	\$199.83	\$319.16
% upside	(23%)	111%	236%
% weight	5%	25%	5%

Blended PT: \$183.8
% Δ: 94%

Risk/Reward:

Base Case

Growth:

MAU Growth	20%	20%	15%	14%	14%	14%	14%	13%	12%	11%	10%	9%	8%	(1.5%)	<- Incremental Growth (Post 2030)
DAU/MAU Conversion	200	200	175	175	200	200	160	145	125	110	100	90	80	74%	<- Mature State GM

Monetization:

First-Year MAU penetration (bps y/y)	100	100	150	150	150	150	125	110	95	80	65	50	35	35%	<- Mature State EBIT - GM Δ
Second-Year MAU penetration (bps y/y)	50	50	140	140	140	140	117	82	57	42	37	32	27		

Bear Case

Growth:

MAU Growth	17%	17%	12%	11%	11%	11%	12%	11%	10%	9%	8%	7%	6%	(2.0%)	<- Incremental Growth (Post 2030)
DAU/MAU Conversion	175	175	150	150	175	175	130	115	95	80	70	60	50	74%	<- Mature State GM

Monetization:

First-Year MAU penetration (bps y/y)	50	50	100	100	100	100	75	60	45	30	15	15	15	57%	<- Mature State EBIT - GM Δ
Second-Year MAU penetration (bps y/y)	25	25	90	90	90	90	67	32	37	27	22	22	22		

Bull Case

Growth:

MAU Growth	23%	23%	18%	17%	17%	17%	16%	15%	14%	13%	12%	11%	10%	(1.5%)	<- Incremental Growth (Post 2030)
DAU/MAU Conversion	275	250	225	225	250	250	210	195	175	160	150	140	130	74%	<- Mature State GM

Monetization:

First-Year MAU penetration (bps y/y)	125	125	175	175	175	175	140	125	110	95	80	65	50	27.5%	<- Mature State EBIT - GM Δ
Second-Year MAU penetration (bps y/y)	75	75	165	165	165	165	132	97	72	57	52	47	42		

Appendix 2: DCF

DCF

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Sensitivity Toggles	
Revenue Growth	-
GM %	-
EBIT %	-
Mature State Growth	2.0%
Duolingo WACC	9.1%

		Historical			Estimates																		
Calendar Year		12/30/19	12/30/20	12/30/21	12/30/22	12/30/23	12/30/24	12/30/25	12/30/26	12/30/27	12/30/28	12/30/29	12/30/30	12/30/31	12/30/32	12/30/33	12/30/34	12/30/35	12/30/36	12/30/37	12/30/38	12/30/39	12/30/40
Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
Fiscal Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E
Revenue		71	162	251	368	527	714	927	1,139	1,359	1,581	1,790	1,992	2,186	2,366	2,526	2,659	2,759	2,821	2,877	2,935	2,993	3,053
% y/y			129%	55.1%	46.6%	43.5%	35.4%	29.8%	23.0%	19.3%	16.3%	13.2%	11.3%	9.8%	8.3%	6.8%	5.3%	3.8%	2.3%	0.8%	(0.7%)	(2.2%)	(3.7%)
post-toggle % y/y					46.6%	43.5%	35.4%	29.8%	23.0%	19.3%	16.3%	13.2%	11.3%	9.8%	8.3%	6.8%	5.3%	3.8%	2.3%	2.0%	2.0%	2.0%	2.0%
Gross Profit		50	116	182	269	389	528	684	842	1,004	1,168	1,323	1,473	1,616	1,750	1,869	1,967	2,041	2,087	2,129	2,172	2,215	2,259
% GM		-	-	-	73.2%	73.7%	74.0%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%
post-toggle % GM					73.2%	73.7%	74.0%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%	74.0%
EBIT		(8)	3	(4)	8	33	91	165	241	325	414	500	587	679	773	865	953	1,032	1,100	1,122	1,145	1,167	1,191
% EBIT		-	-	-	2.3%	6.3%	12.8%	17.8%	21.2%	23.9%	26.2%	27.9%	29.5%	31.1%	32.7%	34.2%	35.8%	37.4%	39.0%	39.0%	39.0%	39.0%	39.0%
post-toggle % EBIT					2.3%	6.3%	12.8%	17.8%	21.2%	23.9%	26.2%	27.9%	29.5%	31.1%	32.7%	34.2%	35.8%	37.4%	39.0%	39.0%	39.0%	39.0%	39.0%
delta % GM & % EBIT					70.9%	67.4%	61.2%	56.1%	52.7%	49.9%	47.7%	46.0%	44.4%	42.9%	41.3%	39.7%	38.1%	36.6%	35.0%	35.0%	35.0%	35.0%	35.0%
(-) Taxes		-	(0)	(0)	(0)	(7)	(19)	(35)	(51)	(69)	(87)	(105)	(124)	(140)	(159)	(177)	(194)	(209)	(221)	(224)	(228)	(231)	(234)
% tax rate		-	1%	(4%)	4%	21%	21%	21%	21%	21%	21%	21%	21%	21%	21%	20%	20%	20%	20%	20%	20%	20%	20%
NOPAT		(8)	3	(4)	8	26	72	130	190	257	327	394	463	539	614	688	759	823	879	898	917	937	957
(+) Reinvestment		7	13	25	14	16	45	46	55	60	67	74	78	54	47	37	26	14	-	-	-	-	-
% revenue		10%	8%	10%	4%	3%	6%	5%	5%	4%	4%	4%	4%	2%	2%	1%	1%	0%	-	-	-	-	-
UFCF		(1)	16	21	22	42	117	176	245	317	393	469	541	593	661	726	785	837	879	898	917	937	957
Discount Rate					1.0	1.1	1.2	1.3	1.4	1.6	1.7	1.9	2.0	2.2	2.4	2.7	2.9	3.2	3.4	3.8	4.1	4.5	4.9
PV FCF					22	38	96	133	169	201	229	250	264	266	271	273	271	265	255	239	224	210	199

Mature State (2035+)

2040E+
Incremental Decel
(1.5%)

% GM
74.0%

% GM & % EBIT Delta
35.0%

Eventual Tax Rate
21.0%

Terminal Value Calculation

Last Year UFCF	957
LT Growth Rate	2.0%
TV FCF	976
TV	13,821
PV of TV	2,837
Phase 1 PV FCF	3,871
Implied EV	6,708
(+) Cash	591
(-) Debt	30
Implied MC	7,269
(/) FDSO	36.4

Implied DCF PT \$199.8
% Δ 111%

Appendix 3: Revenue Build Part 1

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

[illegible]

Appendix 3: Revenue Build Part 2

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Calendar Year	Historical					Estimates (Annual)										Historical (Quarterly)																					
	*****	*****	*****	*****	*****	*****	*****	*****	*****	*****	*****	*****	*****	*****	*****	3/30/18	6/30/18	9/30/18	*****	3/30/19	6/30/19	9/30/19	*****	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	*****	3/30/22	6/30/22	9/30/22	#		
Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2030	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2021	2022	2022	2022	2022
Fiscal Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2030	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2021	2022	2022	2022	2022
Fiscal Quarter	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q3'		
Fiscal Date	Units	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'18	Q2'18	Q3'18	Q4'18	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22			
Incremental English Learners					8.8	12.3																		2	17	5	12	17	(9)	30	11	13	43	29			
% q/q																																					
% y/y						38	24																														
Incremental Learners					37.7	23.6																		11	50	34	56	43	(18)	56	13	27	95	54			
% q/q																																					
% y/y							(37%)																														
% Incrementally English					23.3%	52.3%																		16.6%	34.0%	14.1%	20.5%	39.1%	50.1%	54.2%	85.4%	48.2%	45.5%	54.5%			
bps q/q																																					
bps y/y						2,907																															
% q/q																																					
% y/y						38	(14)																														
Average MAUs (BoP)		17.9	20.3	23.3	27.3	36.7	47.8	55.2	57.0	65.0	73.4	82.2	91.3	100.4	109.5	20.3	21.5	23.0	24.4	25.9	26.9	26.6	28.2	27.5	33.5	39.2	37.0	36.9	39.8	37.9	41.7	42.4	49.2	49.5			
New MAUs		14.4	16.6	19.6	24.9	28.4	39.1	42.5	43.9	48.7	54.0	59.2	64.6	69.8	74.7	4.3	4.8	4.9	5.2	4.8	3.6	5.4	3.5	9.1	9.5	2.2	4.1	8.7	4.3	9.8	7.3	13.6	8.2	8.4			
% of MAU		80%	82%	84%	91%	77%	82%	77%	77%	75%	74%	72%	71%	70%	68%	85%	89%	85%	86%	74%	53%	81%	49%	133%	113%	22%	45%	94%	44%	103%	70%	128%	66%	68%			
% y/y																																					
Churned MAUs		(12.0)	(13.6)	(15.6)	(15.6)	(24.6)	(30.6)	(35.3)	(35.9)	(40.3)	(45.2)	(50.2)	(55.5)	(60.8)	(66.0)	(3.1)	(3.3)	(3.5)	(3.7)	(3.8)	(3.9)	(3.9)	(4.1)	(3.1)	(3.8)	(4.4)	(4.2)	(5.8)	(6.3)	(6.0)	(6.6)	(6.8)	(7.9)	(7.9)			
% of MAU		67%	67%	67%	57%	67%	(64%)	(64%)	(63%)	(62%)	(62%)	(61%)	(61%)	(61%)	(60%)	(15.2%)	(15.2%)	(15.2%)	(15.2%)	(14.5%)	(14.5%)	(14.5%)	(14.5%)	(11.3%)	(11.3%)	(11.3%)	(11.3%)	(15.7%)	(15.7%)	(15.7%)	(15.7%)	(16.0%)	(16.0%)	(16.0%)			
% y/y																																					

Appendix 3: Revenue Build Part 3

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

[illegible]

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Appendix 3: Revenue Build Part 5

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

[illegible]

Appendix 3: Revenue Build Part 6

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Calendar Year	Historical					Estimates (Annual)										Historical (Quarterly)																		
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2030E	3/30/18	6/30/18	9/30/18	12/30/18	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	
Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2030E	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2030E	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	
Fiscal Date	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	2030E	Q1'18	Q2'18	Q3'18	Q4'18	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	
Units	137.8	70.9	62.4	72.6	72.3	68.8	69.8	69.1	68.4	67.7	67.0	66.4	65.7	65.0	65.0					79.2	82.8	84.0	84.0	84.3	85.5	89.0	85.9	87.6	91.6	83.7	81.8	80.0	79.0	
ARPU																																		
% q/q																					5%	1%	(0%)	0%	1%	4%	(4%)	2%	5%	(9%)	(2%)	(2%)	(1%)	
% y/y		(49%)	(12%)	16%	(0%)	(5%)	1%	(1%)	(1%)	(1%)	(1%)	(1%)	(1%)	(1%)	(1%)								6%	3%	6%	2%	4%	7%	(6%)	(5%)	(9%)	(14%)		
% 2y			(55%)	2%	16%	(5%)	(3%)	0%	(2%)	(2%)	(2%)	(2%)	(2%)	(2%)	(2%)												11%	11%	(0%)	(3%)	(5%)	(8%)		
Subscription	9	23	55	118	181	274	409	553	701	853	1,008	1,162	1,312	1,454	1,454					9.0	12.0	15.4	18.5	22.2	28.3	32.3	34.7	40.1	43.5	46.0	51.1	58.0	65.2	
% q/q																					33%	29%	20%	20%	28%	14%	7%	15%	9%	6%	11%	14%	12%	
% y/y	1,300%	157%	138%	114%	54%	52%	49%	46%	43%	40%	37%	34%	31%	28%	28%								147%	136%	109%	88%	81%	54%	42%	47%	45%	50%		
% 2y		3,500%	512%	410%	229%	133%	126%	102%	72%	54%	44%	36%	30%	25%	25%												346%	263%	198%	177%	162%	131%		
% 3y			8,470%	1,211%	684%	400%	248%	206%	156%	109%	82%	66%	54%	44%	44%																545%	445%		
Minutes per MAU		131	190	247	263	254	262	309	357	404	446	482	496	511	511	41	31	29	29	32	45	57	55	50	62	66	67	71	65	64	63	63	63	
% q/q																	(25.9%)	(5.2%)	0.3%	9.4%	40.2%	27.9%	(3.0%)	(9.0%)	23.8%	5.3%	2.7%	4.9%	(8.0%)	(1.5%)	(1.6%)	-		
% y/y			45%	30%	6%	(3%)	3%	18%	16%	13%	11%	8%	3%	3%	3%				(23%)	46%	97%	90%	58%	40%	15%	22%	40%	4%	(2%)	(6%)	(11%)	(3%)		
Total Minutes		3,054	5,177	9,050	10,622	12,679	14,937	20,093	26,224	33,189	40,708	48,362	54,296	60,398	60,398	888	704	708	754	858	1,188	1,607	1,524	1,689	2,446	2,427	2,488	2,814	2,465	2,670	2,673	3,102	3,120	
% q/q																	(20.7%)	0.6%	6.5%	13.8%	38.5%	35.3%	(5.2%)	10.8%	44.8%	(0.8%)	2.5%	13.1%	(12.4%)	8.3%	0.1%	16.0%	0.6%	
% y/y			70%	75%	17%	19%	18%	35%	31%	27%	23%	19%	12%	11%	11%				(3%)	69%	127%	102%	97%	106%	51%	63%	67%	1%	10%	7%	10%	27%		
Total Ad Minutes		226	383	670	787	939	1,106	1,488	1,943	2,458	3,015	3,582	4,022	4,474	4,474	66	52	52	56	64	88	119	113	125	181	180	184	208	183	198	198	230	231	
% total minutes		7%	7%	7%	7%	7%	7%	7%	7%	7%	7%	7%	7%	7%	7%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	7.4%	
% q/q																	(21%)	1%	6%	14%	38%	35%	(5%)	11%	45%	(1%)	3%	13%	(12%)	8%	0%	16%	1%	
% y/y			70%	75%	17%	19%	18%	15%	12%	9%	6%	3%	(0%)	(3%)	(3%)				(3%)	69%	127%	102%	97%	106%	51%	63%	67%	1%	10%	7%	10%	27%		
\$ Per Minute of Advertisement		\$0.04	\$0.04	\$0.05	\$0.05	\$0.05	\$0.05	\$0.05	\$0.05	\$0.05	\$0.06	\$0.06	\$0.06	\$0.06	\$0.06									\$0.040	\$0.038	\$0.037	\$0.049	\$0.044	\$0.050	\$0.046	\$0.056	\$0.051	\$0.049	
% q/q																									(6.1%)	(0.5%)	31.2%	(9.2%)	11.5%	(8.0%)	23.3%	(9.1%)	(5.1%)	
% y/y			6%	19%	(3%)	(5%)	6%	6%	5%	5%	4%	4%	3%	3%	3%												11%	32%	22%	15%	15%	(2%)		
Advertising \$ Per Free MAU		\$0.56	\$0.79	\$1.01	\$0.97	\$0.98	\$1.25	\$1.56	\$1.88	\$2.21	\$2.52	\$2.72	\$2.91	\$2.91	\$2.91									\$0.15	\$0.18	\$0.19	\$0.26	\$0.24	\$0.25	\$0.23	\$0.28	\$0.25	\$0.24	
% q/q																									17%	5%	35%	(5%)	3%	(9%)	22%	(9%)	(4%)	
% y/y			39%	29%	(4%)	1%	28%	25%	21%	17%	14%	8%	7%	7%	7%												58%	40%	21%	9%	4%	(3%)		
Advertising	-	-	15	28	39	45	50	71	98	131	168	209	243	279	279					-	-	-	-	5.0	6.8	6.7	9.0	9.3	9.1	9.0	11.1	11.7	11.2	
% q/q																									36%	(1%)	34%	3%	(2%)	(0%)	23%	5%	(5%)	
% y/y			85%	40%	16%	12%	42%	38%	33%	28%	24%	16%	15%	15%	15%													85%	33%	34%	23%	27%	24%	

Appendix 3: Revenue Build Part 7

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical					Estimates (Annual)										Historical (Quarterly)																	
Calendar Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	3/30/18	6/30/18	9/30/18	3/30/19	6/30/19	9/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	3/30/22	6/30/22				
Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	
Fiscal Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'		Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	
Fiscal Date	Units	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'18	Q2'18	Q3'18	Q4'18	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	
Applications:																																		
China			15.2		22.4	18.3	16.0	15.0	14.0	13.0	13.0	13.0	13.0	13.0																				
India			4.0		7.1	9.1	9.7	10.3	10.9	11.5	12.0	12.5	13.0	13.4	13.7																			
Canada			2.0		3.2	4.0	4.2	4.5	4.7	5.0	5.2	5.4	5.6	5.8	5.9																			
Pakistan			1.4		2.3	3.2	3.4	3.6	3.8	4.0	4.2	4.3	4.5	4.6	4.7																			
South Korea			0.0		2.5	2.6	2.8	2.9	3.1	3.2	3.4	3.5	3.7	3.8	3.9																			
United Arab Emirates					2.2	2.4	2.6	2.7	2.9	3.0	3.2	3.3	3.4	3.5	3.6																			
Nigeria					1.9	2.1	2.3	2.4	2.5	2.7	2.8	2.9	3.0	3.1	3.2																			
United Kingdom					1.7	2.1	2.2	2.3	2.5	2.6	2.7	2.8	2.9	3.0	3.1																			
Brazil					1.4	1.9	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9																			
Vietnam					1.7	1.7	1.9	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.6																			
International Applicants		48.4	50.6	55.2	46.4	47.4	47.1	49.9	52.7	55.5	58.0	60.4	62.6	64.6	66.3																			
% y/y		4%	4%	9%	(16%)	2%	7%	6%	6%	5%	5%	4%	4%	3%	3%																			
Tests to Applicants Ratio		109x	115x	112x	112x	112x	112x	112x	112x	112x	112x	112x	112x	112x																				
% y/y			5%	(3%)	-	-	-	-	-	-	-	-	-	-																				
# of TOEFLs Sold		2.3	2.3	2.4	2.1	2.1	2.1	2.2	2.3	2.5	2.6	2.7	2.8	2.9	2.9																			
% y/y		-	-	6%	(16%)	2%	(1%)	6%	6%	5%	5%	4%	4%	3%	3%																			
% market share		43.4%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%	39.7%																				
# of IELTS Sold		3.0	3.5	3.7	3.1	3.2	3.2	3.4	3.6	3.7	3.9	4.1	4.2	4.3	4.5																			
% y/y		3.4%	16.7%	6.2%	(15.9%)	2.2%	(0.8%)	6.1%	5.6%	5.1%	4.6%	4.1%	3.6%	3.1%	2.6%																			
% market share		56.6%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%	60.3%																				
Tests Market (IELTS + TOEFLS + DUOL)		5.3	5.8	6.2	5.2	5.3	5.2	5.6	5.9	6.2	6.5	6.7	7.0	7.2	7.4																			
% y/y		1.9%	9.4%	6.2%	(15.9%)	2.2%	(0.8%)	6.1%	5.6%	5.1%	4.6%	4.1%	3.6%	3.1%	2.6%																			
% 2y		9.5%	11.5%	16.2%	(10.8%)	(14.1%)	1.4%	5.3%	12.1%	11.1%	10.0%	9.0%	7.9%	6.9%	5.8%																			
% 3y		18.2%	19.8%	18.4%	(2.3%)	(8.8%)	(14.8%)	7.6%	11.2%	17.9%	16.2%	14.5%	12.9%	11.3%	9.7%																			
# of DUOL Tests Sold				0.02	0.34	0.49	0.66	0.85	1.01	1.19	1.34	1.49	1.65	1.78	1.90									0.02	0.09	0.11	0.08	0.10	0.10	0.13	0.16	0.16	0.16	
% market share				0.3%	6.6%	9.3%	12.6%	15.2%	17.2%	19.2%	20.7%	22.2%	23.7%	24.7%	25.7%																			
% q/q																																		
% y/y				1,924%	43%	34%	28%	20%	17%	13%	12%	11%	7%	7%										511%	22%	(25%)	20%	(4%)	39%	21%	(0%)	(1%)		
\$ Per Test			59	44	50	50	50	50	50	60	60	60	60	60									50	50	50	50	50	50	50	50	50	50		
% y/y				(25%)	13%	-	-	0%	20%	-	-	-	-	-																				

Appendix 3: Revenue Build Part 8

Revenue Build

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical					Estimates (Annual)									Historical (Quarterly)																		
Calendar Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	3/30/18	6/30/18	9/30/18	12/30/18	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	
Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year		2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	
Fiscal Date	Units	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'18	Q2'18	Q3'18	Q4'18	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	
English Test		1	1	1	15	25	33	42	51	71	80	90	99	107	114	-	-	-	-	-	-	-	-	0.8	4.6	5.6	4.2	5.0	4.8	6.7	8.1	8.1	8.0	
% q/q																									511%	22%	(25%)	20%	(4%)	39%	21%	(0%)	(1%)	
% y/y					1,416%	63%	34%	28%	20%	41%	13%	12%	11%	7%	7%													569%	5%	19%	93%	60%	66%	
\$ In-App Purchases Per DAU					\$0.2	\$0.7	\$1.2	\$1.7	\$2.1	\$2.5	\$2.8	\$3.1	\$3.3	\$3.4	\$3.4	-	-	-	-	-	-	-	-	0.03	0.04	0.08	0.10	0.10	0.16	0.19	0.26	0.27	0.30	
% q/q																									40%	113%	23%	8%	49%	21%	41%	2%	10%	
% y/y					195%	71%	35%	25%	20%	15%	10%	5%	3%	2%														296%	323%	140%	173%	160%	92%	
Other (Gems, Ect.)		-	-	-	2	7	16	26	39	56	74	94	112	129	145	-	-	-	-	-	-	-	-	0.2	0.3	0.7	0.8	1.0	1.4	1.8	2.7	3.4	3.9	
% q/q																									84%	100%	24%	22%	42%	30%	45%	27%	16%	
% y/y					248%	130%	65%	50%	42%	34%	26%	19%	15%	13%														452%	327%	179%	226%	240%	179%	
Total Revenue		-	-	71	162	251	368	527	714	927	1,139	1,359	1,581	1,790	1,992	12	16	19	23	28	40	45	48	55	59	64	73	81	88					
% q/q																																		
% y/y					129%	55%	47%	43%	35%	30%	23%	19%	16%	13%	11%		25%	24%	19%	21%	42%	13%	7%	15%	6%	8%	15%	11%	9%					
% 2y					254%	127%	110%	94%	76%	60%	47%	39%	32%	26%									125%	156%	133%	108%	343%	276%	227%	215%	189%	121%		
% 3y						420%	226%	185%	152%	116%	90%	71%	57%	47%																				
Subscription				78%	73%	72%	75%	78%	77%	76%	75%	74%	73%	73%		72%	76%	79%	80%	79%	71%	71%	72%	72%	74%	72%	70%	71%	74%					
Advertising				21%	17%	15%	12%	10%	10%	11%	12%	12%	13%	14%		-	-	-	-	18%	17%	15%	19%	17%	15%	14%	15%	14%	13%					
English				1%	9%	10%	9%	8%	7%	8%	7%	7%	6%	6%		-	-	-	-	3%	11%	12%	9%	9%	8%	11%	11%	10%	9%					
Other				-	1%	3%	4%	5%	6%	6%	7%	7%	7%	7%		-	-	-	-	1%	1%	1%	2%	2%	2%	3%	4%	4%	4%					

Appendix 4: Cost Build Part 1

Fiscal Date	Units	Historical			Estimates (Annual)									Historical (Quarterly)												Estimates (Quarterly)							
		2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23
Subscription Revenue		55	118	181	274	409	553	701	853	1,007	1,162	1,312	1,454	9	12	15	18	22	28	32	35	40	44	46	51	58	65	73	78	88	99	107	115
New Subscriber Revenue		50	99	152	225	325	427	530	635	740	843	942	1,032					19	25	28	29	34	35	38	43	48	56	61	64	71	78	85	92
Returning Subscriber Revenue		5	18	29	49	83	125	171	218	268	319	370	421					3	4	5	5	6	9	8	8	10	9	12	14	17	20	22	23
New Subscriber		0.8	1.4	2.1	3.3	4.7	6.2	7.8	9.4	11.0	12.7	14.3	15.9					0.8	0.9	1.2	1.2	1.4	1.5	1.5	1.8	2.1	2.4	2.8	3.1	3.3	3.6	4.0	4.3
Returning Subscriber		0.1	0.3	0.4	0.7	1.2	1.8	2.5	3.2	4.0	4.8	5.6	6.5					0.1	0.2	0.2	0.3	0.3	0.4	0.4	0.5	0.5	0.5	0.6	0.7	0.9	1.0	1.1	1.2
% Revenue Distribution:																																	
App Store		55.2%	51.3%	50.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%					55.1%	50.0%	50.7%	50.7%	50.1%	51.6%	51.6%	49.0%	51.1%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%	53.5%
Google Play		23.5%	26.9%	29.0%	28.4%	28.4%	28.4%	28.4%	28.4%	28.4%	28.4%	28.4%	28.4%					29.3%	26.5%	25.6%	27.1%	28.3%	29.3%	29.3%	29.0%	29.1%	28.4%	28.4%	28.4%	28.4%	28.4%	28.4%	28.4%
Facebook		10.0%	5.0%	-	-	-	-	-	-	-	-	-	-					-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Stripe		-	5.0%	10.1%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%					-	10.1%	11.0%	11.0%	10.3%	12.1%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%	
Other Platforms		11.3%	11.8%	10.4%	10.1%	10.1%	10.1%	10.1%	10.1%	10.1%	10.1%	10.1%	10.1%					15.6%	13.4%	12.7%	11.2%	11.3%	8.8%	7.0%	14.0%	11.8%	10.1%	10.1%	10.1%	10.1%	10.1%	10.1%	10.1%
Total		100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%					100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
New Subscriber Fees:																																	
App Store Fee (First Year)		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%					30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Google Play Fee (First Year)		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%					30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Facebook Fee (First Year)		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%					30%	30%	30%													

Appendix 4: Cost Build Part 2

Fiscal Date	Units	Historical			Estimates (Annual)									Historical (Quarterly)												Estimates (Quarterly)							
		2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23
In-App Purchases		-	2	7	16	26	39	56	74	94	112	129	145					0	0	1	1	1	1	2	3	3	4	4	5	6	6	7	7
% Revenue Distribution (Adjusted):																																	
App Store		70%	66%	64%	65%	65%	65%	65%	65%	65%	65%	65%	65%		65.3%	65.4%	66.4%	65.2%	63.9%	63.8%	63.8%	62.8%	63.7%	65.3%	65%	65%	65%	65%	65%	65%	65%	65%	
Google Play		30%	34%	36%	35%	35%	35%	35%	35%	35%	35%	35%	35%		34.7%	34.6%	33.6%	34.8%	36.1%	36.2%	36.2%	37.2%	36.3%	34.7%	35%	35%	35%	35%	35%	35%	35%	35%	
% Distribution Fees:																																	
App Store Fee		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Google Play Fee		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
In-App Purchases COGS		(0.6)	(2.1)	(4.8)	(7.9)	(11.8)	(16.7)	(22.3)	(28.1)	(33.5)	(38.6)	(43.4)		(0.1)	(0.1)	(0.2)	(0.2)	(0.3)	(0.4)	(0.6)	(0.8)	(1.0)	(1.2)	(1.2)	(1.2)	(1.2)	(1.4)	(1.7)	(1.9)	(2.0)	(2.2)		
In-App Purchases Gross Profit		1	5	11	18	28	39	52	66	78	90	101		0.1	0.2	0.5	0.6	0.7	1.0	1.3	1.9	2.4	2.8	2.8	2.8	3.2	4.0	4.5	4.7	5.2			
% gross margin		70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%		70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%	70%
Advertising		15	28	39	45	50	71	98	131	168	209	243	279		5	7	7	9	9	9	9	11	12	11	11	11	11	12	12	13	13		
Advertising COGS		(6)	(11)	(15)	(18)	(20)	(29)	(39)	(52)	(67)	(83)	(97)	(112)		(2)	(3)	(3)	(4)	(4)	(4)	(4)	(4)	(5)	(4)	(4)	(4)	(4)	(5)	(5)	(5)	(5)	(5)	
Advertising Gross Profit		9	17	23	27	30	43	59	79	101	125	146	168		3.0	4.1	4.0	5.4	5.6	5.4	5.4	6.7	7.0	6.7	6.6	6.4	7.2	7.2	7.6	8.0			
% gross margin		60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	1	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%
Testing		1	15	25	33	42	51	71	80	90	99	107	114		1	5	6	4	5	5	7	8	8	8	8	8	9	9	10	11	12		
Testing COGS		(0.4)	(6.1)	(9.9)	(13.2)	(16.9)	(20.2)	(28.5)	(32.1)	(35.9)	(39.7)	(42.7)	(45.6)		(0.3)	(1.8)	(2.2)	(1.7)	(2.0)	(1.9)	(2.7)	(3.2)	(3.2)	(3.2)	(3.2)	(3.2)	(3.5)	(3.8)	(4.1)	(4.4)	(4.7)		
Testing Gross Profit		0.6	9.1	14.8	19.8	25.4	30.3	42.7	48.2	53.8	59.5	64.0	68.3		0.5	2.8	3.4	2.5	3.0	2.9	4.0	4.9	4.8	4.8	4.9	4.8	5.2	5.7	6.1	6.6	7.0		
% gross margin		60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%		60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%	60%
New Subscriber COGS		(13.5)	(25.6)	(37.8)	(58.3)	(82.9)	(108.9)	(135.1)	(161.8)	(188.4)	(214.8)	(239.9)	(263.0)		(5.1)	(6.0)	(6.6)	(7.2)	(8.4)	(8.7)	(9.5)	(10.5)	(12.0)	(14.3)	(15.5)	(16.4)	(18.0)	(19.9)	(21.7)	(23.4)			
Paid Subscriber COGS		(0.7)	(2.3)	(3.7)	(5.9)	(11.0)	(16.5)	(22.5)	(28.8)	(35.3)	(42.0)	(48.8)	(55.6)		(0.4)	(0.4)	(0.6)	(0.7)	(0.8)	(1.2)	(1.1)	(1.0)	(1.3)	(1.2)	(1.5)	(1.8)	(2.3)	(2.7)	(2.9)	(3.1)			
Subscriber COGS		(14.4)	(28.3)	(41.8)	(62.9)	(93.9)	(125.4)	(157.7)	(190.6)	(223.8)	(256.8)	(288.8)	(318.6)		(5.9)	(7.1)	(8.0)	(7.3)	(9.0)	(10.2)	(11.2)	(11.5)	(12.5)	(15.0)	(17.1)	(18.3)	(20.3)	(22.6)	(24.6)	(26.5)			
In-App COGS		-	(0.6)	(2.1)	(4.8)	(7.9)	(11.8)	(16.7)	(22.3)	(28.1)	(33.5)	(38.6)	(43.4)		(0.1)	(0.1)	(0.2)	(0.2)	(0.3)	(0.4)	(0.6)	(0.8)	(1.0)	(1.2)	(1.2)	(1.4)	(1.7)	(1.9)	(2.0)	(2.2)			
Advertising COGS		(6.0)	(11.0)	(15.4)	(17.8)	(20.0)	(28.5)	(39.4)	(52.4)	(67.4)	(83.5)	(97.2)	(111.7)		(2.0)	(2.7)	(2.7)	(3.6)	(3.7)	(3.6)	(3.6)	(4.5)	(4.7)	(4.5)	(4.4)	(4.3)	(4.8)	(4.8)	(5.1)	(5.3)			
English Tests COGS		(0.4)	(6.1)	(9.9)	(13.2)	(16.9)	(20.2)	(28.5)	(32.1)	(35.9)	(39.7)	(42.7)	(45.6)		(0.3)	(1.8)	(2.2)	(1.7)	(2.0)	(1.9)	(2.7)	(3.2)	(3.2)	(3.2)	(3.2)	(3.2)	(3.5)	(3.8)	(4.1)	(4.4)	(4.7)		
Total COGS		(21)	(46)	(69)	(98.6)	(138.7)	(186)	(242)	(297)	(355)	(413)	(467)	(519)		(8)	(12)	(13)	(13)	(15)	(16)	(18)	(20)	(21)	(24)	(26)	(27)	(31)	(33)	(36)	(39)			
Total Gross Profit		50	116	182	269	389	528	684	842	1,004	1,168	1,323	1,472		20	28	32	35	40	43	46	53	60	65	70	75	84	94	101	109			
% gross margin		70.7%	71.3%	72.4%	73.2%	73.7%	74.0%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%		70.8%	70.5%	71.1%	73.4%	72.9%	72.6%	71.6%	72.7%	73.5%	73.0%	72.9%	73.2%	73.4%	73.7%	73.8%	73.8%			

Appendix 4: Cost Build Part 3

		Historical			Estimates (Annual)									Historical (Quarterly)												Estimates (Quarterly)							
Fiscal Date	Units	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23
S&M Employees		14	23	36	57	86	116	146	170	188	205	219	229	12	13	13	14	14	15	20	23	25	29	33	36	44	49	49	57	74	80	70	86
% q/q							35.9%	25.9%	15.9%	10.9%	8.9%	6.9%	4.9%	10.0%	-	4.5%	4.3%	4.2%	33.3%	13.1%	10.7%	16.3%	13.6%	9.9%	19.7%	12.5%	-	15.8%	29.6%	8.7%	(12.4%)	22.2%	
% y/y			64%	61%	56%	51%	36%	26%	16%	11%	9%	7%	5%				20%	14%	52%	64%	74%	94%	66%	61%	74%	68%	48%	56%	69%	63%	43%	51%	
Wage (S&M)		119,744	122,405	125,394	132,025	136,788	139,524	142,315	145,161	148,064	151,025	154,046	157,127	118,722	119,155	119,870	121,230	121,983	121,690	122,561	123,383	123,903	124,737	125,555	127,379	129,203	131,026	132,850	135,022	136,309	135,612	136,836	138,397
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%	
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%	3.2%	4.3%	5.0%	5.8%			6.0%	5.5%	3.5%	3.0%	2.5%		
S&M Spend (Wages)		(2)	(2)	(4)	(7)	(11)	(16.2)	(20.8)	(24.6)	(27.8)	(30.9)	(33.7)	(36.0)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.5)	(0.6)	(0.7)	(0.8)	(0.9)	(1.0)	(1.2)	(1.4)	(1.6)	(1.6)	(1.9)	(2.5)	(2.7)	(2.4)	(3.0)
% adj. S&M		11%	6%	7%	10%	11%	14%	15%	15%	15%	15%	14%	14%	16%	16%	10%	8%	8%	5%	6%	7%	5%	9%	7%	8%	10%	11%	9%	9%	12%	13%	9%	10%
% q/q														10.4%	0.6%	5.7%	5.0%	3.9%	34.3%	13.9%	11.2%	17.1%	14.4%	11.5%	21.4%	14.1%	1.4%	17.7%	30.8%	8.2%	(11.6%)	23.6%	
% y/y			41%	76%	69%	61%	53%	28%	18%	13%	11%	9%	7%				23%	16%	55%	67%	77%	99%	70%	66%	81%	77%	57%	65%	78%	69%	47%	55%	
S&M Spend (Implied Fixed/Misc)		(12)	(32)	(49)	(62)	(87)	(100)	(120)	(141)	(162)	(182)	(200)	(215)	(1.9)	(2.1)	(3.8)	(4.7)	(5.0)	(7.8)	(10.4)	(8.7)	(13.4)	(9.1)	(13.5)	(12.7)	(12.9)	(12.8)	(17.5)	(18.7)	(18.1)	(17.9)	(24.5)	(26.2)
% adj. S&M		89%	94%	93%	90%	89%	86%	85%	85%	85%	85%	86%	86%	84%	84%	90%	92%	92%	95%	94%	93%	95%	91%	93%	92%	90%	89%	91%	91%	88%	87%	91%	90%
% q/q														11.0%	79.3%	24.8%	5.7%	57.9%	32.6%	(16.6%)	54.9%	(32.1%)	47.7%	(5.5%)	1.6%	(1.1%)	36.5%	7.2%	(3.2%)	(1.1%)	36.5%	7.2%	
% y/y			156%	53%	27%	40%	15%	20%	18%	15%	13%	10%	8%				162%	273%	176%	84%	170%	16%	30%	47%	(4%)	40%	30%	47%	40%	40%	40%	40%	
Adj. S&M		(14.0)	(34.1)	(52.7)	(68.5)	(97.3)	(115.9)	(140.5)	(165.2)	(189.5)	(212.8)	(233.8)	(251.1)	(2)	(2)	(4)	(5)	(5)	(8)	(11)	(9)	(14)	(10)	(15)	(14)	(14)	(14)	(19)	(21)	(21)	(21)	(27)	(29)
% revenue		(20%)	(21%)	(21%)	(19%)	(18%)	(16%)	(15%)	(15%)	(14%)	(13%)	(13%)	(13%)	(18%)	(16%)	(21%)	(22%)	(19%)	(21%)	(24%)	(19%)	(26%)	(17%)	(23%)	(19%)	(18%)	(16%)	(20%)	(20%)	(18%)	(16%)	(20%)	(20%)
% q/q														11%	67%	23%	6%	54%	33%	(15%)	52%	(29%)	45%	(4%)	3%	0%	33%	8%	(0%)	0%	30%	9%	
% y/y			143%	54%	30%	42%	19%	21%	18%	15%	12%	10%	7%				140%	233%	165%	83%	163%	21%	32%	48%	1%	44%	32%	48%	44%	43%	41%	41%	

Appendix 4: Cost Build Part 4

Fiscal Date	Units	Historical			Estimates (Annual)									Historical (Quarterly)											Estimates (Quarterly)									
		2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23	
Research / Learning Employees		23	45	49	53	58	63	67	71	74	77	80	81	16	22	22	23	24	25	40	45	45	51	51	49	49	49	49	49	53	53	47	47	58
% q/q							7.9%	6.9%	5.9%	4.9%	3.9%	2.9%	1.9%	37.5%	-	4.5%	4.3%	4.2%	60.0%	12.5%	-	13.3%	-	(3.9%)	-	-	-	-	8.9%	-	(11.8%)	-	23.4%	
% y/y			96%	9%	9%	9%	8%	7%	6%	5%	4%	3%	2%				50%	14%	82%	96%	88%	104%	28%	9%	9%	(4%)	(4%)	9%	9%	(4%)	(4%)	9%		
Software Engineers		115	175	221	265	292	321	352	383	416	449	483	517	106	110	110	115	120	125	165	175	182	199	213	221	245	276	276	265	294	303	303	292	
% q/q							10.0%	9.5%	9.0%	8.5%	8.0%	7.5%	7.0%	3.8%	-	4.5%	4.3%	4.2%	31.8%	6.1%	4.1%	9.4%	7.1%	3.7%	10.8%	12.5%	-	(3.7%)	10.8%	3.1%	-	(3.7%)		
% y/y			52%	27%	20%	10%	10%	10%	9%	9%	8%	8%	7%				13%	14%	50%	52%	52%	59%	29%	27%	35%	38%	29%	20%	20%	10%	10%	10%		
Wage (Research/Learning)		122,067	124,779	127,826	134,587	139,442	142,231	145,076	147,977	150,937	153,956	157,035	160,175	121,025	121,466	122,195	123,582	124,350	124,051	124,939	125,777	126,307	127,158	127,991	129,850	131,709	133,568	135,428	137,641	138,953	138,243	139,490	141,082	
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%	
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%				2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%	3.2%	4.3%	5.0%	5.8%	6.0%	5.5%	3.5%	3.0%	2.5%		
Wage (Software Engineers)		117,758	120,375	123,314	129,836	134,520	137,211	139,955	142,754	145,609	148,521	151,491	154,521	116,753	117,179	117,882	119,219	119,961	119,672	120,529	121,337	121,848	122,669	123,473	125,267	127,060	128,854	130,647	132,783	134,048	133,363	134,566	136,102	
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%	
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%				2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%	3.2%	4.3%	5.0%	5.8%	6.0%	5.5%	3.5%	3.0%	2.5%		
R&D Spend (Wages)		(16)	(22)	(31)	(41)	(47)	(53)	(59)	(65)	(72)	(79)	(86)	(93)	(3.6)	(3.9)	(3.9)	(4.1)	(4.3)	(4.5)	(6.2)	(6.7)	(7.0)	(7.7)	(8.2)	(8.5)	(9.4)	(10.5)	(10.7)	(10.6)	(11.7)	(11.7)	(11.8)	(12.0)	
% adj. R&D		52%	43%	35%	33%	29%	27%	25%	25%	24%	24%	24%	24%	60%	55%	47%	47%	48%	39%	41%	47%	38%	37%	32%	33%	36%	38%	30%	31%	34%	33%	26%	26%	
% q/q														8.7%	0.6%	5.7%	5.0%	3.9%	37.6%	8.1%	3.6%	11.0%	6.3%	3.7%	10.3%	11.9%	1.4%	(0.1%)	9.9%	0.3%	0.9%	1.2%		
% y/y			40%	44%	31%	15%	12%	11%	11%	10%	10%	9%	8%				21%	16%	59%	62%	60%	71%	32%	27%	35%	36%	30%	25%	25%	12%	11%	13%		
R&D Spend (Implied Fixed/Misc)		(14)	(28)	(59)	(82)	(115)	(144)	(173)	(199)	(224)	(249)	(274)	(299)	(2.4)	(3.2)	(4.3)	(4.6)	(4.8)	(7.1)	(9.0)	(7.6)	(11.1)	(13.1)	(17.5)	(17.2)	(16.4)	(17.3)	(24.4)	(24.1)	(23.0)	(24.2)	(34.2)	(33.7)	
% adj. R&D		48%	57%	65%	67%	71%	73%	75%	75%	76%	76%	76%	76%	40%	45%	53%	53%	52%	61%	59%	53%	62%	63%	68%	67%	64%	62%	70%	69%	66%	67%	74%	74%	
% q/q														35.0%	36.1%	6.3%	3.8%	47.6%	27.7%	(15.8%)	46.8%	17.6%	33.2%	(1.5%)	(4.7%)	5.6%	41.1%	(1.5%)	(4.7%)	5.6%	41.1%	(1.5%)		
% y/y			96%	107%	40%	40%	25%	20%	15%	13%	11%	10%	9%				103%	122%	108%	65%	133%	86%	94%	126%	47%	32%	40%	40%	40%	40%	40%	40%		
Adj. R&D		(30.0)	(50.3)	(90.3)	(123.4)	(162.4)	(197)	(232)	(264)	(296)	(328)	(360)	(391)	(6)	(7)	(8)	(9)	(9)	(12)	(15)	(14)	(18)	(21)	(26)	(26)	(26)	(28)	(35)	(35)	(35)	(36)	(46)	(46)	
% revenue		(42%)	(31%)	(36%)	(34%)	(31%)	(28%)	(25%)	(23%)	(22%)	(21%)	(20%)	(20%)	(48%)	(45%)	(42%)	(38%)	(32%)	(29%)	(34%)	(30%)	(33%)	(35%)	(40%)	(35%)	(32%)	(31%)	(37%)	(34%)	(30%)	(28%)	(33%)	(31%)	
% q/q														19%	17%	6%	4%	27%	32%	(6%)	27%	15%	23%	0%	0%	8%	26%	(1%)	(0%)	4%	28%	(1%)		
% y/y			67%	80%	37%	32%	21%	18%	14%	12%	11%	10%	9%				54%	64%	85%	64%	98%	80%	69%	80%	42%	34%	37%	35%	34%	29%	31%	32%		

Appendix 4: Cost Build Part 5

Fiscal Date	Units	Historical				Estimates (Annual)								Historical (Quarterly)										Estimates (Quarterly)									
		2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23
People & Leadership		9	21	47	68	85	105	127	151	176	203	229	254	6	9	9	9	10	10	16	21	26	33	40	47	57	67	67	68	74	88	84	85
% q/q							23.0%	21.0%	19.0%	17.0%	15.0%	13.0%	11.0%		46.7%	-	4.5%	4.3%	4.2%	64.3%	28.3%	22.7%	26.7%	22.2%	17.0%	21.9%	17.9%	-	0.9%	9.3%	17.9%	(3.8%)	0.9%
% y/y			129%	122%	45%	25%	23%	21%	19%	17%	15%	13%	11%					60%	14%	87%	129%	170%	228%	144%	122%	121%	106%	68%	45%	30%	30%	25%	25%
Product Management		-	10	40	58	72	89	108	128	150	172	195	216	6	-	-	-	-	-	5	10	16	23	31	40	52	61	61	58	67	80	77	72
% q/q							23.0%	21.0%	19.0%	17.0%	15.0%	13.0%	11.0%	(100.0%)							116.0%	58.9%	48.6%	36.4%	26.8%	29.6%	18.4%	-	(5.5%)	16.2%	18.4%	(3.8%)	(5.5%)
% y/y				308%	45%	25%	23%	21%	19%	17%	15%	13%	11%				(100%)					595%	308%	233%	165%	95%	45%	30%	30%	25%	25%		
Design		35	44	53	60	66	72	78	84	89	93	97	100	34	33	33	35	36	38	44	44	46	47	51	53	54	61	61	60	63	70	67	66
% q/q							9.0%	8.0%	7.0%	6.0%	5.0%	4.0%	3.0%	(2.9%)	-	4.5%	4.3%	4.2%	18.1%	0.2%	2.8%	3.9%	7.0%	3.6%	3.7%	12.5%	-	(1.4%)	3.7%	12.5%	(4.3%)	(1.4%)	
% y/y			29%	18%	15%	10%	9%	8%	7%	6%	5%	4%	3%				6%	14%	34%	29%	27%	26%	14%	18%	19%	29%	21%	15%	15%	15%	10%	10%	
Operations		35	42	44	46	49	51	53	55	56	58	59	60	16	33	33	35	36	38	43	42	42	43	44	44	49	49	46	46	51	51	49	
% q/q							4.5%	4.0%	3.5%	3.0%	2.5%	2.0%	1.5%	106.3%	-	4.5%	4.3%	4.2%	15.6%	(2.4%)	0.1%	0.4%	3.6%	0.1%	(1.2%)	12.5%	-	(5.5%)	(1.2%)	12.5%	-	(5.5%)	
% y/y			23%	4%	5%	5%	5%	4%	4%	3%	3%	2%	2%				125%	14%	31%	23%	18%	13%	2%	4%	3%	15%	11%	5%	5%	5%	5%	5%	
Wage (People & Leadership)		79,696	81,467	83,456	87,870	91,040	92,861	94,718	96,612	98,545	100,515	102,526	104,576	79,015	79,304	79,780	80,685	81,186	80,991	81,571	82,118	82,464	83,019	83,564	84,777	85,991	87,205	88,419	89,864	90,721	90,257	91,071	92,111
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%				2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%		3.2%	4.3%	5.0%	5.8%	6.0%	5.5%	3.5%	3.0%	2.5%
Wage (Product Management)		115,137	117,695	120,569	126,945	131,525	134,156	136,839	139,576	142,367	145,215	148,119	151,081	114,154	114,570	115,258	116,565	117,290	117,008	117,846	118,636	119,135	119,938	120,724	122,478	124,231	125,985	127,738	129,827	131,064	130,394	131,571	133,072
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%				2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%		3.2%	4.3%	5.0%	5.8%	6.0%	5.5%	3.5%	3.0%	2.5%
Wage (Design)		110,200	112,649	115,400	121,503	125,887	128,404	130,972	133,592	136,264	138,989	141,769	144,604	109,260	109,658	110,316	111,568	112,261	111,992	112,793	113,550	114,028	114,796	115,549	117,227	118,905	120,584	122,262	124,261	125,445	124,804	125,930	127,367
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%				2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%		3.2%	4.3%	5.0%	5.8%	6.0%	5.5%	3.5%	3.0%	2.5%
Wage (Operations)		108,265	110,671	113,373	119,369	123,676	126,149	128,672	131,246	133,871	136,548	139,279	142,065	107,341	107,732	108,379	109,609	110,290	110,025	110,813	111,556	112,025	112,780	113,520	115,168	116,817	118,466	120,115	122,078	123,242	122,612	123,718	125,130
% q/q							2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	0.4%	0.6%	1.1%	0.6%	(0.2%)	0.7%	0.7%	0.4%	0.7%	0.7%	1.5%	1.4%	1.4%	1.4%	1.4%	1.6%	1.0%	(0.5%)	0.9%	1.1%
% y/y			2%	2%	5%	4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%				2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%		3.2%	4.3%	5.0%	5.8%	6.0%	5.5%	3.5%	3.0%	2.5%
G&A Spend (Wages)		(8)	(11)	(17)	(26)	(32)	(37.3)	(43.8)	(50.8)	(58.3)	(66.2)	(74.2)	(82.1)	(1.6)	(2.0)	(2.0)	(2.1)	(2.2)	(2.3)	(2.9)	(3.2)	(3.5)	(3.9)	(4.5)	(5.0)	(5.7)	(6.7)	(6.8)	(6.7)	(7.3)	(8.3)	(8.1)	(8.0)
% adj. G&A		55%	38%	43%	38%	33%	30%	30%	30%	30%	31%	32%	34%	58%	60%	58%	47%	35%	38%	44%	35%	49%	47%	41%	38%	37%	38%	41%	36%	33%	33%	35%	31%
% q/q														19.4%	0.6%	5.7%	5.0%	3.9%	27.7%	8.4%	10.1%	12.9%	14.5%	11.6%	14.0%	16.9%	1.4%	(1.4%)	8.5%	14.9%	(2.4%)	(1.8%)	
% y/y			37%	60%	53%	23%	18%	17%	16%	15%	14%	12%	11%				33%	16%	47%	51%	58%	72%	54%	59%	64%	70%	51%	33%	27%	25%	20%	19%	
G&A Spend (Implied Fixed/Misc)		(6)	(17)	(23)	(43)	(64)	(87)	(104)	(121)	(135)	(147)	(156)	(161)	(1.2)	(1.3)	(1.4)	(2.3)	(4.1)	(3.7)	(3.7)	(5.9)	(3.6)	(4.5)	(6.6)	(8.1)	(9.6)	(11.1)	(9.9)	(12.1)	(14.4)	(16.7)	(14.8)	(18.2)
% adj. G&A		45%	62%	57%	62%	67%	70%	70%	70%	70%	69%	68%	66%	42%	40%	42%	53%	65%	62%	56%	65%	51%	53%	59%	62%	63%	62%	59%	64%	67%	67%	65%	69%
% q/q														9.2%	7.8%	64.0%	74.2%	(10.1%)	0.5%	58.6%	(38.0%)	23.8%	46.2%	22.9%	18.8%	15.7%	(11.3%)	22.9%	18.8%	15.7%	(11.3%)	22.9%	
% y/y			174%	32%	87%	50%	35.0%	20.0%	16.0%	12.0%	9.0%	6.0%	3.0%				236%	177%	158%	150%	(11%)	22%	78%	38%	164%	147%		50%	50%	50%	50%	50%	50%
Adj. G&A		(14.0)	(27.9)	(39.7)	(68.7)	(95.9)	(123.9)	(147.6)	(171.3)	(193.3)	(213.3)	(230.2)	(242.8)	(3)	(3)	(3)	(4)	(6)	(6)	(7)	(9)	(7)	(8)	(11)	(13)	(15)	(18)	(17)	(19)	(22)	(25)	(23)	(26)
% revenue		(20%)	(17%)	(16%)	(19%)	(18%)	(17%)	(16%)	(15%)	(14%)	(13%)	(13%)	(12%)	(23%)	(21%)	(18%)	(19%)	(22%)	(15%)	(15%)	(19%)	(13%)	(14%)	(17%)	(18%)	(20%)	(17%)	(18%)	(19%)	(20%)	(17%)	(18%)	
% q/q														15.1%	3.5%	30.2%	41.6%	(5.2%)	10.9%	36.5%	(21.2%)	18.5%	31.5%	18.3%	17.0%	16.2%	(6.5%)	13.0%	15.2%	15.5%	(8.3%)	14.1%	
% y/y			99%	42%	73%	40%	29%	19%	16%	13%	10%	8%	5%					120%	81%	94%	103%	13%	41%	67%	45%	116%	111%	50%	44%	41%	40%	38%	39%
Adj. EBIT		(8.0)	3.5	(1.1)	8.4	33.1	91.2	164.6	241.4	325.3	413.9	499.6	587.2	(11.1)	(12.9)	(15.8)	(18.3)	(0.9)	2.4	(0.7)	2.7	0.9	3.4	(5.8)	0.3	4.3	4.5	(1.1)	0.8	7.5	12.1	5.6	7.9
% margin		(11.3%)	2.1%	(0.5%)	2.3%	6.3%	12.8%	17.8%	21.2%	23.9%	26.2%	27.9%	29.5%	(88%)	(82%)	(81%)	(79%)	(3%)	6%	(1%)	6%	2%	6%	(9%)	0%	5%	5%	(1%)	1%	7%	10%	4%	5%
% q/q														16.4%	23.0%	15.7%		(94.9%)	(354.1%)	(127.9%)	(509.5%)	(66.2%)	269.5%	(270.8%)	(105.8%)	#####	4.8%	(123.9%)	(173.2%)	865.2%	60.7%	(53.7%)	41.5%
% y/y			(143%)	(133%)	(840%)	293%	175%	81%	47%		27%	21%	18%				(92%)	(118%)	(96%)	(115%)	(198%)	43%	774%	(88%)	367%	32%		(81%)	133%	77%	172%	(625%)	915%
Street					6.6	21.3	52.5	96.7	170.9	207.3																							

Appendix 5: Income Statement Part 1

Income Statement

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical			Estimates (Annual)									Historical (Quarterly)													
Calendar Year		#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	3/31/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22
Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'
Fiscal Date	Units	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22
Revenue	\$m	71	162	251	368	527	714	927	1,139	1,359	1,581	1,790	1,992	12.5	15.6	19.4	23.2	28.1	40.0	45.3	48.3	55.4	58.8	63.6	73.0	81.2	88.4
COGS	\$m	(21)	(46)	(69)	(99)	(139)	(186)	(242)	(297)	(355)	(413)	(467)	(519)	(3.7)	(4.6)	(5.7)	(6.7)	(8.2)	(11.8)	(13.1)	(12.9)	(15.0)	(16.1)	(18.1)	(20.0)	(21.5)	(23.9)
Gross Profit	\$m	50	116	182	269	389	528	684	842	1,004	1,168	1,323	1,473	8.7	11.0	13.8	16.5	19.9	28.2	32.2	35.4	40.3	42.7	45.5	53.1	59.7	64.5
S&M	\$m	(15)	(35)	(59)	(30)	-								(2.6)	(2.7)	(4.3)	(5.3)	(5.5)	(8.6)	(11.1)	(9.7)	(19.8)	(9.6)	(15.3)	(14.5)	(14.9)	(15.3)
Adj S&M	\$m	(14)	(34)	(53)	(68)	(97)	(116)	(140)	(165)	(190)	(213)	(234)	(251)	(2.2)	(2.5)	(4.2)	(5.1)	(5.4)	(8.3)	(11.0)	(9.4)	(14.2)	(10.0)	(14.5)	(13.9)	(14.3)	(14.4)
R&D	\$m	(32)	(53)	(104)	(64)	-								(6.3)	(7.4)	(8.6)	(9.3)	(9.6)	(12.1)	(15.9)	(15.4)	(22.5)	(21.9)	(29.3)	(30.0)	(29.8)	(34.2)
Adj R&D	\$m	(30)	(50)	(90)	(123)	(162)	(197)	(232)	(264)	(296)	(328)	(360)	(391)	(5.9)	(7.1)	(8.2)	(8.7)	(9.1)	(11.6)	(15.2)	(14.3)	(18.1)	(20.8)	(25.7)	(25.7)	(25.8)	(27.8)
G&A	\$m	(16)	(44)	(79)										(3.4)	(3.8)	(4.0)	(5.2)	(7.3)	(7.4)	(8.2)	(20.8)	(11.5)	(11.6)	(29.6)	(25.9)	(26.9)	(30.1)
Adj G&A	\$m	(14)	(28)	(40)	(69)	(96)	(124)	(148)	(171)	(193)	(213)	(230)	(243)	(2.9)	(3.3)	(3.4)	(4.4)	(6.3)	(6.0)	(6.6)	(9.0)	(7.1)	(8.4)	(11.1)	(13.1)	(15.3)	(17.8)
EBIT	\$m	(14)	(16)	(60)	(27)	-	-	-	-	-	-	-	-	(3.5)	(2.9)	(3.2)	(4.5)	(2.5)	0.1	(3.1)	(10.6)	(13.4)	(0.5)	(28.7)	(17.4)	(11.8)	(15.0)
Adj EBIT	\$m	(8)	3	(4)	8	33	91	165	241	325	414	500	587	(2.3)	(1.8)	(2.1)	(1.8)	(0.9)	2.4	(0.7)	2.7	0.9	3.4	(5.7)	(2.6)	4.3	4.5
(+) D&A	\$m	1	2	3	4	7	35	46	56	67	78	88	98	0	0	0	0	0	1	1	1	1	1	1	1	1	1
(+) SBC	\$m	4	17	41	84	121	114	129	136	135	126	107	79	1	1	1	1	1	2	2	13	3	3	21	15	15	18
Adj. EBITDA	\$m	(9)	3	(16)	62	161	240	340	434	528	618	695	765	(2.3)	(1.8)	(2.0)	(3.0)	(0.9)	2.4	(0.8)	2.5	(10.3)	3.1	(7.3)	(2.0)	3.5	4.3
Other Exp.	\$m	(1)	(0)	(0)	0	1	1	1	1	2	2	2	3	(0.2)	(0.2)	(0.1)	(0.1)	(0.2)	0.0	0.1	(0.2)	0.0	(0.3)	0.2	(0.0)	0.3	(0.1)
Adj. Income Before Tax	\$m	(14)	(16)	(60)	(27)	34	92	166	243	327	416	502	590	(3.4)	(2.8)	(3.0)	(4.4)	(2.2)	0.1	(3.2)	(10.4)	(13.5)	(0.2)	(28.9)	(17.4)	(12.1)	(14.9)
Taxes	\$m	-	(0)	(0)	(0)	(7)	(19)	(35)	(51)	(69)	(87)	(105)	(124)	-	-	-	-	(0.0)	0.0	(0.0)	(0.0)	(0.0)	(0.0)	(0.1)	(0.1)	(0.0)	(0.1)
GAAP Net Income	\$m	(14)	(16)	(60)	(27)	27	73	131	192	258	329	397	466	(3.4)	(2.8)	(3.0)	(4.4)	(2.2)	0.0	(3.2)	(10.4)	(13.5)	(0.2)	(29.0)	(17.5)	(12.2)	(15.0)
Non-GAAP Net Income	\$m	(9)	4	(9)	8	27	73	131	192	258	329	397	466	(3)	(1.7)	(1.9)	(1.7)	(0.7)	2.3	(0.8)	2.9	0.9	3.5	(10.5)	(3.3)	4.1	4.4
WDSO	m#	-	14.4	43.6	39.8	41.1	42	44	45	47	48	49	50						16.2	14.4	14.4	13.8	13.2	29.5	43.6	43.1	39.3
Diluted EPS	\$	\$0.00	(\$1.07)	(\$2.49)	(\$0.69)	\$0.66	\$1.72	\$3.00	\$4.26	\$5.55	\$6.87	\$8.10	\$9.35						-	(0.25)	(0.82)	(1.04)	(0.01)	(0.98)	(0.46)	(0.31)	(0.38)
Adj. Diluted EPS	\$	\$0.00	\$0.33	(\$0.05)	\$0.19	\$0.66	\$1.72	\$3.00	\$4.26	\$5.55	\$6.87	\$8.10	\$9.35						0.17	(0.06)	0.23	0.06	0.19	(0.20)	(0.09)	0.09	0.10

Appendix 5: Income Statement Part 2

Income Statement

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical			Estimates (Annual)									Historical (Quarterly)													
Calendar Year		#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	3/31/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22
Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'
Fiscal Date	Units	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22
Margin Check																											
COGS	%	(29.3%)	(28.4%)	(27.6%)	(26.8%)	(26.3%)	(26.0%)	(26.1%)	(26.1%)	(26.1%)	(26.1%)	(26.1%)	(26.1%)	(30%)	(29%)	(29%)	(29%)	(29%)	(30%)	(29%)	(27%)	(27%)	(27%)	(28%)	(27%)	(26%)	(27%)
GM	%	70.7%	71.6%	72.4%	73.2%	73.7%	74.0%	73.9%	73.9%	73.9%	73.9%	73.9%	73.9%	70.0%	70.6%	70.9%	71.0%	70.8%	70.5%	71.1%	73.4%	72.9%	72.6%	71.6%	72.7%	73.5%	73.0%
S&M	%	(21%)	(22%)	(24%)	(8%)	-	-	-	-	-	-	-	-	(21%)	(17%)	(22%)	(23%)	(20%)	(22%)	(25%)	(20%)	(36%)	(16%)	(24%)	(20%)	(18%)	(17%)
Adj S&M	%	(19.8%)	(21.1%)	(21.0%)	(18.6%)	(18.4%)	(16.2%)	(15.2%)	(14.5%)	(13.9%)	(13.5%)	(13.1%)	(12.6%)	(18%)	(16%)	(21%)	(22%)	(19%)	(21%)	(24%)	(19%)	(26%)	(17%)	(23%)	(19%)	(18%)	(16%)
R&D	%	(44.6%)	(32.8%)	(41.4%)	(17.4%)	-	-	-	-	-	-	-	-	(50%)	(47%)	(44%)	(40%)	(34%)	(30%)	(35%)	(32%)	(41%)	(37%)	(46%)	(41%)	(37%)	(39%)
Adj R&D	%	(42.4%)	(31.1%)	(36.0%)	(33.6%)	(30.8%)	(27.6%)	(25.0%)	(23.2%)	(21.8%)	(20.7%)	(20.1%)	(19.7%)	(48%)	(45%)	(42%)	(38%)	(32%)	(29%)	(34%)	(30%)	(33%)	(35%)	(40%)	(35%)	(32%)	(31%)
Adj G&A	%	(19.8%)	(17.3%)	(15.8%)	(18.7%)	(18.2%)	(17.4%)	(15.9%)	(15.0%)	(14.2%)	(13.5%)	(12.9%)	(12.2%)	(23%)	(21%)	(18%)	(19%)	(22%)	(15%)	(15%)	(19%)	(13%)	(14%)	(17%)	(18%)	(19%)	(20%)
Adj EBIT	%	(11.3%)	2.1%	(1.6%)	2.3%	6.3%	12.8%	17.8%	21.2%	23.9%	26.2%	27.9%	29.5%	(18%)	(12%)	(11%)	(8%)	(3%)	6%	(1%)	6%	2%	6%	(9%)	(4%)	5%	5%
D&A	%	1.8%	1.4%	1.1%	1.2%	1.2%	4.9%	4.9%	4.9%	4.9%	4.9%	4.9%	4.9%	3%	2%	1%	1%	1%	2%	1%	1%	1%	1%	1%	1%	1%	1%
SBC	%	5.3%	10.5%	16.3%	22.9%	23.0%	16.0%	14.0%	12.0%	10.0%	8.0%	6.0%	4.0%	6%	5%	5%	5%	4%	4%	4%	26%	5%	5%	32%	20%	18%	20%
Adj. EBITDA	%	(12.9%)	2.0%	(6.6%)	16.8%	30.5%	33.7%	36.7%	38.1%	38.8%	39.1%	38.8%	38.4%	(18%)	(11%)	(10%)	(13%)	(3%)	6%	(2%)	5%	(19%)	5%	(11%)	(3%)	4%	5%
Other Exp.	%	(0.8%)	(0.2%)	(0.0%)	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	(1%)	(1%)	(1%)	(0%)	(1%)	0%	0%	(0%)	0%	(1%)	0%	(0%)	0%	(0%)
Adj. Income Before Tax	%	(19.2%)	(9.7%)	(23.9%)	(7.3%)	6.4%	12.9%	17.9%	21.3%	24.1%	26.3%	28.0%	29.6%	(27%)	(18%)	(16%)	(19%)	(8%)	0%	(7%)	(22%)	(24%)	(0%)	(45%)	(24%)	(15%)	(17%)
Taxes	%	-	0.3%	0.3%	1.2%	(21.0%)	(21%)	(21%)	(21%)	(21%)	(21%)	(21%)	(21%)	-	-	-	-	0%	22%	1%	0%	0%	1%	0%	1%	0%	1%
GAAP Net Income	%	(19.2%)	(9.8%)	(24.0%)	(7.4%)	5.1%	10.2%	14.1%	16.8%	19.0%	20.8%	22.1%	23.4%	(27%)	(18%)	(16%)	(19%)	(8%)	0%	(7%)	(22%)	(24%)	(0%)	(46%)	(24%)	(15%)	(17%)
Non-GAAP Net Income	%	(12.2%)	2.3%	(3.8%)	2.3%	5.1%	10.2%	14.1%	16.8%	19.0%	20.8%	22.1%	23.4%	(27%)	(11%)	(10%)	(7%)	(3%)	6%	(2%)	6%	2%	6%	(17%)	(4%)	5%	5%
Δ GM & EBIT %					71%	67%	61%	56%	53%	50%	48%	46%	44%														
% y/y growth																											
Revenue	%	129%	55%	47%	43%	35%	30%	23%	19%	16%	13%	11%		125%	156%	133%	108%	97%	47%	40%	51%	47%	50%				
COGS	%	122%	50%	43%	41%	34%	30%	23%	19%	16%	13%	11%		119%	156%	132%	91%	83%	37%	38%	55%	43%	48%				
GM	%	131%	57%	48%	44%	36%	30%	23%	19%	16%	13%	11%		128%	155%	134%	115%	103%	51%	41%	50%	48%	51%				
Adj S&M	%	143%	54%	30%	42%	19%	21%	18%	15%	12%	10%	7%		140%	233%	165%	83%	163%	21%	32%	48%	1%	44%				
Adj R&D	%	67%	80%	37%	32%	21%	18%	14%	12%	11%	10%	9%		54%	64%	85%	64%	98%	80%	69%	80%	42%	34%				
Adj G&A	%	99%	42%	73%	40%	29%	19%	16%	13%	10%	8%	5%		120%	81%	94%	103%	13%	41%	67%	45%	116%	111%				
Adj EBIT	%	(143%)	(217%)	(308%)	293%	175%	81%	47%	35%	27%	21%	18%		(60%)	(229%)	(68%)	(247%)	(198%)	44%	773%	(197%)	367%	31%				

Appendix 5: Income Statement Part 3

Income Statement

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical			Estimates (Annual)									Historical (Quarterly)													
Calendar Year		#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	3/31/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22
Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'
Fiscal Date	Units	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22
Non-GAAP Net Income	%		(143%)	(356%)	(188%)	220%	172%	80%	47%	35%	27%	21%	18%					(79%)	(240%)	(60%)	(266%)	(221%)	51%	1,274%	(215%)	376%	26%
WDSO	%			202%	(9%)	3%	3%	3%	3%	3%	3%	2%	2%										(19%)	106%	202%	212%	198%
Adj. Diluted EPS	%			(114%)	(508%)	246%	161%	74%	42%	30%	24%	18%	16%										13%	227%	(140%)	70%	(47%)
% q/q growth																											
Revenue	%														25%	24%	19%	21%	42%	13%	7%	15%	6%	8%	15%	11%	9%
COGS	%														23%	23%	19%	22%	44%	11%	(2%)	17%	7%	12%	10%	8%	11%
GM	%														26%	25%	20%	21%	42%	14%	10%	14%	6%	7%	17%	13%	8%
Adj S&M	%														11%	67%	23%	6%	54%	33%	(15%)	52%	(29%)	45%	(4%)	3%	0%
Adj R&D	%														19%	17%	6%	4%	27%	32%	(6%)	27%	15%	23%	0%	0%	8%
Adj G&A	%														15%	3%	30%	42%	(5%)	11%	36%	(21%)	18%	31%	18%	17%	16%
Adj EBIT	%														(21%)	13%	(11%)	(49%)	(354%)	(128%)	(509%)	(66%)	273%	(269%)	(54%)	(262%)	5%
Non-GAAP Net Income	%														(51%)	16%	(11%)	(59%)	(428%)	(133%)	(472%)	(70%)	311%	(401%)	(69%)	(224%)	9%
WDSO	%																		(11%)	1%	(4%)	(5%)	124%	48%	(1%)	(9%)	
Adj. Diluted EPS	%																		(137%)	(467%)	(76%)	243%	(206%)	(55%)	(204%)	7%	

Appendix 6: Balance Sheet

Balance Sheet

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical						Estimates (Annual)										Historical (Quarterly)													
Calendar Year		12/30/16	12/30/17	12/30/18	12/30/19	12/30/20	12/30/21	12/30/22	12/30/23	12/30/24	12/30/25	12/30/26	12/30/27	12/30/28	12/30/29	12/30/30	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	
Balance Sheet																															
Cash & ST investments	\$m	-	-	-	60	120	554	632	796	1,027	1,334	1,717	2,171	2,692	3,270	3,892	30	29	33	60	63	81	82	120	117	115	549	554	577	591	
AR	\$m	-	-	-	10	20	33	45	64	87	113	139	166	193	219	243	6	8	7	10	13	17	18	20	23	25	21	33	29	28	
Other Current Assets	\$m	-	-	-	9	17	32	47	71	96	125	153	183	213	241	268	6	7	6	9	11	14	13	17	18	23	26	32	35	36	
Total Current Assets	\$m	-	-	-	79	158	619	724	932	1,211	1,572	2,009	2,520	3,098	3,730	4,404	41.5	43.1	46.7	78.9	87.6	112.5	113.7	158.4	159.0	162.6	596.3	619.3	640.6	655.2	
PPE, Net	\$m	-	-	-	6	9	27	24	34	20	2	(20)	(46)	(77)	(111)	(150)	4	4	5	6	8	8	7	9	10	11	12	27	16	20	
Other Assets	\$m	-	-	-	10	9	29	34	49	66	86	106	127	147	167	185	6	6	6	10	9	9	9	9	8	8	9	29	28	26	
Total Assets	\$m	-	-	-	95	176	661	782	1,014	1,297	1,660	2,096	2,601	3,169	3,785	4,440	51.3	53.2	58.5	95.1	104.8	129.7	130.3	175.7	176.9	181.7	617.8	661.3	684.5	701.1	
AP	\$m	-	-	-	4	2	8	2	3	4	5	7	8	9	10	11	1	2	5	4	3	6	5	2	4	4	9	8	2	2	
Deferred Revenue	\$m	-	-	-	26	55	98	155	228	311	399	489	583	678	772	863	13	15	19	26	35	45	46	55	65	71	80	98	119	128	
Other Current Liabilities	\$m	-	-	-	2	9	13	20	30	42	56	71	88	106	123	141	0	1	1	2	3	4	6	9	15	10	9	13	15	18	
Total Current Liabilities	\$m	-	-	-	32	66	119	177	261	357	460	567	679	792	905	1,015	14.5	18.2	24.8	32.1	41.9	55.1	56.8	65.7	84.1	85.1	98.6	119.1	135.8	147.3	
LT Debt	\$m	-	-	-	9	8	29	26	26	26	26	26	26	26	26	26	7	7	7	9	10	10	10	8	8	8	8	29	28	26	
Total Liabilities	\$m	-	-	-	41	74	148	203	287	383	486	593	704	818	931	1,040	21.1	24.7	31.8	41.3	51.9	64.9	66.3	73.8	92.1	92.9	107.0	148.3	163.8	172.8	
Equity		-	-	-	54	102	513	580	728	914	1,175	1,503	1,897	2,351	2,854	3,399	30	28	27	54	53	65	64	102	85	89	511	513	521	528	
Total Liabilities & Equity	\$m	-	-	-	95	176	661	782	1,014	1,297	1,660	2,096	2,601	3,169	3,785	4,440	51.3	53.2	58.5	95.1	104.8	129.7	130.3	175.7	176.9	181.7	617.8	661.3	684.5	701.1	
Check		-	-	-	-	-	-	-	-	(0)	(0)	(0)	(0)	(0)	(0)	(0)	-	-	(0)	-	-	-	-	-	-	-	-	-	-	-	

Appendix 7: Cash Flow Statement

		Historical						Estimates (Annual)										Historical (Quarterly)													
Calendar Year		12/30/16	12/30/17	12/30/18	12/30/19	12/30/20	12/30/21	12/30/22	12/30/23	12/30/24	12/30/25	12/30/26	12/30/27	12/30/28	12/30/29	12/30/30	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	
Cash Flow Statement																															
NI	\$m	-	-	-	(14)	(16)	(60)	(27)	27	73	131	192	258	329	397	466	(3)	(3)	(3)	(4)	(2)	0	(3)	(10)	(13)	(0)	(29)	(18)	(12)	(15)	
D&A	\$m	-	-	-	1	2	3	4	7	35	46	56	67	78	88	98	0	0	0	0	0	1	1	1	1	1	1	1	1	1	
SBC	\$m	-	-	-	4	17	41	84	121	114	129	136	135	126	107	79	1	1	1	1	1	2	2	13	3	3	21	15	15	18	
Other Non-Cash Expenses	\$m	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	
Δ NWC:																															
Δ in AR	\$m	-	-	-	(5)	(10)	(13)	(11)	(20)	(23)	(26)	(26)	(27)	(27)	(26)	(25)	(2)	(1)	1	(3)	(3)	(4)	(0)	(3)	(3)	(2)	4	(12)	5	0	
Δ in AP	\$m	-	-	-	2	(2)	6	(6)	1	1	1	1	1	1	1	1	(0)	1	2	(1)	(0)	3	(1)	(2)	2	0	4	(1)	(6)	(1)	
Δ in Deferred Revenue	\$m	-	-	-	17	28	43	57	73	83	87	91	93	95	94	91	4	2	4	7	9	10	1	9	10	6	9	18	21	9	
Δ in Other Assets/Liabilities	\$m	-	-	-	(5)	(2)	(11)	(20)	(29)	(30)	(34)	(33)	(34)	(33)	(30)	(28)	(0)	(1)	(1)	(3)	0	(2)	2	(3)	6	(9)	(6)	(2)	(2)	(0.5)	
CFO	\$m	-	-	-	2	18	9	81	180	253	334	417	495	569	631	682	(0.4)	(0.4)	4.0	(1.1)	4.7	8.5	0.8	3.7	5.1	(0.9)	4.5	0.4	20.6	12.7	
Capex	\$m	-	-	-	(4)	(3)	(4)	(10)	(16)	(21)	(28)	(34)	(41)	(48)	(54)	(60)	(0)	(0)	(1)	(2)	(2)	(1)	(0)	(1)	(1)	(1)	(1)	(1)	(1)	(2)	
Other Investing Activities	\$m	-	-	-	6	(1)	(3)	(3)	-	-	-	-	-	-	-	-	6	(0)	1	(0)	(0)	(0)	(0)	(1)	(1)	(1)	(0)	(1)	(1)	(1)	
CFI	\$m	-	-	-	2	(4)	(6)	(13)	(16)	(21)	(28)	(34)	(41)	(48)	(54)	(60)	5.7	(0.6)	(0.1)	(2.5)	(1.7)	(0.6)	(0.4)	(1.3)	(1.8)	(1.9)	(1.5)	(1.1)	(2.4)	(3.3)	
Stock Issuance (Preferred/Comr	\$m	-	-	-	30	45	429	-	-	-	-	-	-	-	-	-	0	0	0	30	0	10	0	35	0	0	426	3	-	-	
Other Financing Activities	\$m	-	-	-	1	2	1	10	-	-	-	-	-	-	-	-	0	0	0	0	0	0	1	1	(6)	(0)	6	2	5	4	
CFF	\$m	-	-	-	31	47	430	10	-	-	-	-	-	-	-	-	0.1	0.2	0.3	30.3	0.3	10.1	0.7	35.8	(6.4)	(0.1)	431.8	5.2	5.2	4.4	
Cash (Beginning)	\$m	-	-	-	24	60	120	554	632	796	1,027	1,334	1,717	2,171	2,692	3,270	24	30	29	33	60	63	81	82	120	117	115	549	554	577	
Δ Cash	\$m	-	-	-	36	61	433	78	164	231	307	383	454	521	578	622	5	(1)	4	27	3	18	1	38	(3)	(3)	435	4	23	14	
Cash (Ending)	\$m	-	-	-	60	120	554	632	796	1,027	1,334	1,717	2,171	2,692	3,270	3,892	30	29	33	60	63	81	82	120	117	115	549	554	577	591	
Foreign Currency Impact																															
Check (BS & CF)		-	-	-	-	-	-	(0)	(0)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	(0)	(0)	(0)	-	-	

Appendix 8: BS/CFS Drivers

Balance Sheet

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical						Estimates (Annual)										Historical (Quarterly)													
Calendar Year		12/30/16	12/30/17	12/30/18	12/30/19	12/30/20	12/30/21	12/30/22	12/30/23	12/30/24	12/30/25	12/30/26	12/30/27	12/30/28	12/30/29	12/30/30	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2019	2019	2019	2019	2020	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	
Drivers																															
AR Days					51.6	46.2	48.3	44.3	44.6	44.6	44.6	44.6	44.6	44.6	44.6	44.6	45.5	44.2	33.1	39.3	42.5	39.8	36.1	38.7	38.5	39.0	29.8	41.4	32.1	29.1	
Other Current Assets (% Revenue)					12.8%	10.8%	12.8%	12.8%	13.5%	13.5%	13.5%	13.5%	13.5%	13.5%	13.5%	13.5%	11.2%	10.5%	8.3%	9.8%	10.1%	8.7%	7.4%	9.0%	8.2%	9.7%	10.3%	11.0%	10.7%	10.1%	
Other Assets (% Revenue)					13.7%	5.3%	11.7%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	9.3%	12.0%	9.6%	8.2%	10.4%	8.4%	5.9%	5.0%	4.5%	3.6%	3.3%	3.6%	10.0%	8.7%	7.4%	
AP Days					67	17	41	8	8	8	8	8	8	8	8	8	33	48	76	52	37	47	32	16	23	24	44	36	9	6	
Deferred Revenue (% Subscription Revenue)					48%	47%	54%	57%	56%	56%	57%	57%	58%	58%	59%	59.3%	35%	31%	31%	36%	40%	39%	36%	39%	41%	41%	44%	48%	51%	49%	
Other Current Liabilities (% Revenue)					2.8%	5.4%	5.2%	5.3%	5.7%	5.9%	6.1%	6.3%	6.5%	6.7%	6.9%	7.1%	0.9%	1.1%	1.1%	2.1%	3.1%	2.7%	3.4%	4.5%	6.8%	4.2%	3.7%	4.5%	4.5%	5.0%	
D&A (% Revenue)					1.8%	1.4%	1.1%	1.2%	1.2%	1.2%	1.2%	1.2%	1.2%	1.2%	1.2%	1%	0.8%	0.5%	0.3%	0.3%	0.4%	0.4%	0.3%	0.3%	0.3%	0.3%	0.3%	0.3%	0.2%	0.3%	
SBC (% Revenue)					5.3%	10.5%	16.3%	22.9%	23.0%	16.0%	14.0%	12.0%	10.0%	8.0%	6.0%	4.0%	1.6%	1.3%	1.1%	1.3%	1.0%	1.1%	0.9%	6.5%	1.2%	1.2%	8.1%	5.0%	4.5%	5.1%	
Capex (% Revenue)					(5.3%)	(2.1%)	(1.4%)	(2.8%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(0.2%)	(0.2%)	(1.5%)	(2.5%)	(1.4%)	(0.4%)	(0.2%)	(0.4%)	(0.4%)	(0.5%)	(0.4%)	(0.2%)	(0.4%)	(0.5%)	
DSO Build																															
SBC		-	-	-	4	17	41	84	121	114	129	136	135	126	107	79	1	1	1	1	1	2	2	13	3	3	21	15	15	18	
Buybacks		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Issuances		-	-	-	30	45	429	-	-	-	-	-	-	-	-	-	0	0	0	30	0	10	0	35	0	0	426	3	-	-	
(/) Stock Price								94.87	94.87	94.87	94.87	94.87	94.87	94.87	94.87	94.87															
Δ in DSO								0.5	1.3	1.2	1.4	1.4	1.4	1.3	1.1	0.8															
Beginning DSO								39.3	39.8	41.1	42.3	43.7	45.1	46.5	47.9	49.0							16	14	14	14	13	30	44	43	
Ending DSO								39.8	41.1	42.3	43.7	45.1	46.5	47.9	49.0	49.8						16	14	14	14	13	30	44	43	39	
Deferred Revenue																															
Subscription Revenue		0.6	9.0	23.0	54.8	117.5	180.7																								
% Subscription Bookings		80.0%	80.0%	80.0%	76.1%	81.4%	80.5%										-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Subscription Bookings		0.8	11.2	28.8	72.1	144.4	224.5	-	-	-	-	-	-	-	-	-	12.7	14.3	19.6	25.6	30.7	36.6	33.8	43.3	50.5	48.9	55.4	69.8	78.5	74.1	
% Total Bookings		80.0%	80.0%	80.0%	81.9%	75.9%	76.3%										78.4%	79.5%	83.1%	84.4%	83.3%	73.8%	72.4%	75.9%	76.7%	75.8%	75.8%	76.8%	77.0%	76.0%	
Total Bookings		1.0	14.0	36.0	88.0	190.2	294.2	-	-	-	-	-	-	-	-	-	16.2	18.0	23.6	30.3	36.9	49.6	46.7	57.1	65.8	64.5	73.1	90.8	102.1	97.5	

Appendix 9: Comps Part 1

Comps

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Calendar Year	Year	Fiscal Year	Fiscal Quarter	Fiscal Date	Units	Historical							
						2014	2015	2016	2017	2018	2019	2020	2021
						2014	2015	2016	2017	2018	2019	2020	2021
						Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
						FY'14	FY'15	FY'16	FY'17	FY'18	FY'19	FY'20	FY'21
Tinder													
Non-Tinder Subscribers						3,694	4,143	4,002	3,946	3,898	4,057	4,477	
MAUs (Match Group)						55,217	63,500	70,500	73,000	78,500	83,000	92,000	
Match Group Revenue						1,020	1,223	1,331	1,730	2,051	2,391	2,983	
Tinder Revenue						51	173	402	805	1,152	1,350	1,485	
Tinder as % revenue						5.0%	14.2%	30.2%	46.5%	56.2%	56.5%	49.8%	
Tinder as % subscribers						10.6%	23.4%	37.5%	49.9%	58.0%	61.0%	60.5%	
Tinder MAUs (Revenue Approach)						2,746	8,986	21,324	33,971	44,088	46,869	45,808	
Tinder MAUs (Subscriber Approach)						5,869	14,837	26,430	36,403	45,523	50,658	55,629	
Tinder Plus Subscribers						439	1,263	2,078	1,710	1,589	1,768	1,949	
Tinder Gold Subscribers						-	-	322	2,215	3,792	4,586	4,899	
Total Subscribers						439	1,263	2,400	3,925	5,381	6,354	6,848	
% Penetration (Revenue Approach)						16.0%	14.1%	11.3%	11.6%	12.2%	13.6%	14.9%	
% Penetration (Subscriber Approach)						7.5%	8.5%	9.1%	10.8%	11.8%	12.5%	12.3%	
Average subscribers - Tinder Plus						439	1,263	2,078	1,710	1,589	1,768	1,949	
Tinder Plus monthly ARPU						9.2	10.8	12.9	12.7	12.4	10.7	14.3	
Tinder Plus Revenue						48	163	321	261	237	227	335	
Average subscribers - Tinder Gold								322	2,215	3,792	4,586	4,899	
Tinder Gold monthly ARPU								20.5	20.5	20.1	20.4	19.6	
Tinder Gold Revenue								79	544	915	1,124	1,150	

Appendix 9: Comps Part 2

Comps

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical							
Calendar Year		#####	#####	#####	#####	#####	#####	#####	#####
Year		2014	2015	2016	2017	2018	2019	2020	2021
Fiscal Year		2014	2015	2016	2017	2018	2019	2020	2021
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	FY'14	FY'15	FY'16	FY'17	FY'18	FY'19	FY'20	FY'21

Bumble

Bumble MAU							12,300		
Bumble Paid Subscribers (US)						668	857	1,068	
Bumble Paid Subscribers (International)						187	285	413	
Total Paid Subscribers						856	1,142	1,482	
% penetration							9.3%		
Average Monthly ARPU						26.8	26.3	30.0	
Bumble Revenue						276	360	533	
Badoo MAU							28,400		
Badoo Subscribers						1,195	1,363	1,394	
% penetration							4.8%		
Average Monthly ARPU						13.8	12.7	13.2	
Badoo Revenue						197	208	220	
Average Monthly Advertising Per Free User							\$0.52		
Badoo Advertising Revenue						16	14	13	

Appendix 9: Comps Part 3

Comps		NASDAQ: DUOL FYE: December 31st Unit in M																					
																Estimates							
Calendar Year		3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	9/30/22	12/30/22	3/30/23	6/30/23	9/30/23	12/30/23	3/30/24	
Year		2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022	2022	2023	2023	2023	2023	2024	
Fiscal Year		2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022	2022	2023	2023	2023	2023	2024	
Fiscal Quarter		Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	
Fiscal Date	Units	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22	Q4'22	Q1'23	Q2'23	Q3'23	Q4'23	Q1'24	
bps YoY										-86	-203	-221	-227	-62	20	-1	-13	-5	-9	0	-3	4	
bps QoQ							42	29	17	-174	-75	11	11	-9	6	-10	0	0	2	-1	-3	6	
Rovio																							
(also owns a bunch of mobile games, eg. Angry Birds)																							
% Weighting		89.69%	88.98%	88.88%	86.55%	84.97%	78.86%	82.57%	72.41%	39.38%	27.99%	33.93%	41.24%	43.70%	40.98%	51.48%	51.80%	50.45%	49.69%	49.71%	46.30%	44.05%	
Games Rev		66,300	65,400	66,400	66,700	62,700	66,900	64,200	64,400	64,900	66,400	68,600	76,500	82,303	75,800	74,673	77,293	77,574	80,877	82,871	83,994	82,156	
MARPPU		39	40	38	38	40	45	43	40	41	43	40	41	42	45	38	37	38	42	45	48	48	
MAU		49,300	45,300	40,700	37,300	36,500	32,500	29,400	27,300	25,600	26,000	32,000	44,800	53,300	45,800	67,153	72,173	71,435	68,169	64,914	61,413	60,332	
DAU		7,000	6,600	5,900	5,900	5,400	5,100	4,500	4,400	4,200	4,200	4,900	6,500	7,300	6,700	9,197	9,766	9,743	9,466	9,152	8,760	8,629	
DAU/MAU		14.20%	14.57%	14.50%	15.82%	14.79%	15.69%	15.31%	16.12%	16.41%	16.15%	15.31%	14.51%	13.70%	14.63%	13.70%	13.53%	13.64%	13.89%	14.10%	14.26%	14.30%	
bps YoY						60	112	81	30	161	46	1	(161)	(271)	(153)	-162	-98	-6	-74	40	73	66	
bps QoQ			37	-7	132	-102	90	-39	81	29	-25	-84	-80	-81	93	-93	-16	11	25	21	17	4	
MPU		1,722	1,619	1,761	1,742	1,583	1,497	1,511	1,614	1,579	1,562	1,706	1,870	1,983	1,696	1,970	2,106	2,049	1,939	1,826	1,756	1,697	
MPU/DAU		24.60%	24.53%	29.85%	29.52%	29.32%	29.35%	33.57%	36.68%	37.60%	37.20%	34.83%	28.78%	27.17%	25.31%	21.42%	21.57%	21.03%	20.48%	19.95%	20.05%	19.67%	
bps YoY						472	482	372	717	828	785	126	(791)	(1,043)	(1,189)	-1341	-721	-613	-483	-146	-152	-137	
bps QoQ			-7	532	-33	-20	2	422	311	91	-40	-237	-605	-161	-186	-389	15	-54	-55	-53	9	-38	
MPU/MAU		3.49%	3.57%	4.33%	4.67%	4.34%	4.61%	5.14%	5.91%	6.17%	6.01%	5.33%	4.18%	3.72%	3.70%	2.93%	2.92%	2.87%	2.84%	2.81%	2.86%	2.81%	
bps YoY						84	103	81	124	183	140	19	-174	-245	-231	-240	-126	-85	-86	-12	-6	-6	
bps QoQ			8	75	34	(33)	27	53	77	26	(16)	(68)	(116)	(45)	(2)	-77	-1	-5	-2	-3	5	-5	

Appendix 9: Industry Part 1

		Historical					
Calendar Year		12/30/16	12/30/17	12/30/18	12/30/19	12/30/20	12/30/21
Year		2016	2017	2018	2019	2020	2021
Fiscal Year		2016	2017	2018	2019	2020	2021
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A
Spending							
Games	\$m	116,980	139,468	166,991	200,096	252,240	
Social Networking		16,712	19,695	21,763	31,395	39,594	
Entertainment		10,452	14,084	15,081	21,487	26,982	
Photo & Video		3,870	5,952	8,406	12,524	16,042	
Lifestyle		6,093	8,352	8,779	8,299	10,049	
Music		6,200	7,719	8,031	8,466	9,995	
Productivity		3,723	4,905	5,543	7,154	8,699	
Books & Reference		2,259	2,732	3,265	5,416	7,331	
Education		2,234	2,899	3,546	5,074	6,480	
Health & Fitness		2,097	3,051	3,891	5,294	6,593	
Utilities		2,261	2,751	3,120	3,853	4,575	
Sports		1,509	1,925	2,531	2,887	3,472	
Business		637	878	1,038	1,823	2,504	
News & Magazines		480	572	696	923	1,165	
Navigation		465	483	557	777	994	
Finance		457	537	563	764	945	
Shopping		203	284	359	602	821	
Food & Drink		328	381	448	664	858	
Weather		282	396	420	566	708	
Medical		211	203	226	283	382	
Travel		186	203	245	214	296	
Total App Spending		177,639	217,470	255,499	318,561	400,725	
% y/y			22.4%	17.5%	24.7%	25.8%	

Appendix 9: Industry Part 2

Industry

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical					
Calendar Year		12/30/16	12/30/17	12/30/18	12/30/19	12/30/20	12/30/21
Year		2016	2017	2018	2019	2020	2021
Fiscal Year		2016	2017	2018	2019	2020	2021
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A
\$ / Download							
Games			2.06	2.15	2.25	2.04	2.17
Photo & Video			0.39	0.52	0.60	0.64	0.69
Utilities			0.20	0.21	0.23	0.24	0.27
Entertainment			1.71	1.96	1.87	2.15	2.41
Social Networking			3.20	3.13	2.99	3.39	3.85
Music			1.42	1.58	1.50	1.27	1.35
Productivity			1.13	1.30	1.28	1.23	1.28
Education			1.51	1.60	1.63	1.52	1.52
Lifestyle			2.61	3.13	2.85	2.28	2.31
Business			0.91	1.05	0.91	0.85	0.84
Health & Fitness			2.06	2.27	2.38	2.12	2.06
Books & Reference			2.93	3.50	3.28	3.29	3.36
Finance			0.57	0.58	0.47	0.42	0.41
Food & Drink			0.59	0.57	0.56	0.46	0.43
Travel			0.35	0.30	0.26	0.16	0.16
Sports			2.59	2.72	2.95	2.17	2.01
News & Magazines			1.41	1.49	1.42	1.18	1.14
Shopping			0.70	0.81	0.68	0.80	0.86
Weather			0.78	1.18	1.14	1.11	1.12
Navigation			1.24	1.48	1.51	1.59	1.67
Medical			1.69	1.44	1.52	1.33	1.44
Average			1.43	1.57	1.54	1.44	1.49
% y/y				9.6%	(2.1%)	(6.3%)	3.7%

Appendix 9: Industry Part 3

Industry

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

Calendar Year	Year	Fiscal Year	Fiscal Quarter	Fiscal Date	Units	Historical					
						12/30/16	12/30/17	12/30/18	12/30/19	12/30/20	12/30/21
						2016	2017	2018	2019	2020	2021
						2016	2017	2018	2019	2020	2021
						Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
						2016A	2017A	2018A	2019A	2020A	2021A
						% y/y		9.6%	(2.1%)	(6.3%)	3.7%

% y/y:

Games	4.5%	4.8%	(9.4%)	6.6%
Photo & Video	31.2%	15.9%	6.4%	7.5%
Utilities	6.6%	6.3%	5.6%	11.2%
Entertainment	14.3%	(4.6%)	14.7%	12.4%
Social Networking	(2.1%)	(4.5%)	13.4%	13.5%
Music	11.5%	(5.3%)	(14.9%)	6.0%
Productivity	15.0%	(1.8%)	(4.2%)	4.1%
Education	6.0%	1.5%	(6.5%)	(0.1%)
Lifestyle	19.8%	(8.9%)	(20.0%)	1.4%
Business	15.3%	(13.5%)	(6.2%)	(1.5%)
Health & Fitness	9.9%	5.2%	(10.9%)	(3.0%)
Books & Reference	19.4%	(6.3%)	0.1%	2.4%
Finance	0.5%	(19.2%)	(10.5%)	(1.0%)
Food & Drink	(3.7%)	(2.1%)	(16.9%)	(7.9%)
Travel	(15.9%)	(13.5%)	(37.1%)	(3.0%)
Sports	5.0%	8.4%	(26.3%)	(7.6%)
News & Magazines	5.8%	(5.2%)	(16.3%)	(3.6%)
Shopping	16.5%	(16.5%)	17.8%	7.6%
Weather	50.8%	(3.2%)	(2.9%)	0.5%
Navigation	18.5%	2.2%	5.2%	5.2%
Medical	(14.8%)	5.7%	(12.6%)	8.8%
Average	10.2%	(2.6%)	(6.3%)	2.8%
conditional cutoff	5.0%	5.0%	5.0%	5.0%

Appendix 10: Demand Part 1

Demand

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical					Estimates (Annual)									
Calendar Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E

US Language Market

Motivated Learners

For Work:

Method #1: Outbound Business Flights Internationally

# of Outbound Business Int. Flyers	6.0	6.3	6.6	1.4	1.3	3.6	4.9	5.7	6.2	6.6	7.0	7.3	7.7	8.0
% 2019 levels				22%	20%	55%	75%	88%	94%	100%	106%	112%	117%	122%
% y/y		6%	4%	(78%)	(8%)	176%	36%	17%	7%	6%	6%	5%	5%	4%
% Servicable	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%	75%
Total DUOL Business SAM	4.5	4.7	4.9	1.1	1.0	2.7	3.7	4.3	4.6	4.9	5.2	5.5	5.8	6.0
% y/y		6%	4%	(78%)	(8%)	176%	36%	17%	7%	6%	6%	5%	5%	4%

Appendix 10: Demand Part 2

Demand

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical						Estimates (Annual)								
Calendar Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Method #2: # of Visas																
Lawful Permanent Resident (BoP)		13.0	13.4	13.6	13.5	13.4	13.1	13.1	13.2	13.3	13.5	13.6	13.6	13.6	13.5	13.5
(+) Immigrant Visas Issued		0.6	0.6	0.5	0.5	0.2	0.3	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.4
% LPR		4.8%	4.2%	3.9%	3.4%	1.8%	2.2%	4.1%	4.1%	4.1%	4.1%	3.0%	3.0%	3.0%	3.0%	3.0%
(-) Implied Churn		(0.2)	(0.4)	(0.7)	(0.6)	(0.5)	(0.3)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)
% LPR		(1.5%)	(2.9%)	(4.8%)	(4.3%)	(3.7%)	(2.5%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.2%)	(3.2%)	(3.2%)	(3.2%)	(3.2%)
Lawful Permanent Resident (EoP)		13.4	13.6	13.5	13.4	13.1	13.1	13.2	13.3	13.5	13.6	13.6	13.6	13.5	13.5	13.5
% immigrant population		30.7%	30.5%	30.0%	29.7%	29.7%	28.8%	28.4%	28.1%	27.7%	27.3%	26.7%	26.0%	25.4%	24.7%	24.1%
% y/y			1.3%	(0.9%)	(0.8%)	(1.9%)	(0.3%)	1.1%	1.1%	1.1%	1.1%	(0.2%)	(0.2%)	(0.2%)	(0.2%)	(0.2%)
Naturalized Citizens (BoP)			19.8	20.4	21.8	21.6	21.5	21.9	22.3	22.6	22.8	23.1	23.4	23.6	23.9	24.1
Citizen Applications	0.97	0.99	0.81	0.83	0.99	0.79	0.95	0.90	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85
(+) Citizenships Granted		0.7	0.8	0.8	0.6	0.8	0.8	0.8	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
% Previous Year Applications		73%	77%	104%	76%	83%	84.7%	84.7%	84.7%	84.7%	84.7%	84.7%	84.7%	84.7%	84.7%	84.7%
% Naturalized Citizens		3.6%	3.7%	3.9%	2.9%	3.8%	3.7%	3.4%	3.2%	3.2%	3.1%	3.1%	3.0%	3.0%	3.0%	3.0%
(-) Implied Churn		(0.2)	0.7	(1.1)	(0.7)	(0.4)	(0.4)	(0.4)	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)
% Naturalized Citizens		(1.0%)	3.4%	(5.0%)	(3.2%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)
Natural Citizens (EoP)		19.8	20.4	21.8	21.6	21.5	21.9	22.3	22.6	22.8	23.1	23.4	23.6	23.9	24.1	24.3
% immigrant population		45.4%	45.7%	48.6%	48.0%	48.7%	48.3%	48.0%	47.5%	46.9%	46.3%	45.8%	45.3%	44.7%	44.1%	43.5%
% y/y			3%	7%	(1%)	(0%)	2%	1.7%	1.4%	1.2%	1.2%	1.1%	1.1%	1.0%	1.0%	1.0%
Unauthorized Immigrants (EoP)		7.9	7.8	6.4	6.0	6.0	6.0	5.8	5.6	5.4	5.2	5.0	4.8	4.7	4.5	4.4
% immigrant population		18.1%	17.5%	14.2%	13.3%	13.6%	13.3%	12.5%	11.8%	11.1%	10.4%	9.8%	9.3%	8.8%	8.3%	7.8%
% y/y			(1%)	(18%)	(6%)	0%	-	(3.5%)	(3.5%)	(3.5%)	(3.5%)	(3.5%)	(3.5%)	(3.5%)	(3.5%)	(3.5%)

Appendix 10: Demand Part 3

Demand

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical						Estimates (Annual)								
Calendar Year		#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####	#####
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Temporary Workers		1.10	1.28	1.40	1.62	2.16	1.95	2.42	2.90	3.43	3.99	4.61	5.27	5.97	6.70	7.48
% nonimmigrants		48%	50%	50%	51%	53%	55%	56%	56%	57%	57%	58%	58%	59%	59%	60%
Students		0.87	0.94	1.01	1.10	1.32	1.07	1.30	1.52	1.75	1.98	2.23	2.48	2.73	2.99	3.25
% nonimmigrants		38%	36%	36%	34%	32%	30%	30%	29%	29%	28%	28%	27%	27%	26%	26%
Exchange Visitors		0.24	0.26	0.28	0.35	0.42	0.37	0.45	0.53	0.63	0.72	0.83	0.94	1.05	1.17	1.30
% nonimmigrants		10%	10%	10%	11%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
Diplomats/Other Representatives		0.09	0.10	0.10	0.12	0.15	0.13	0.16	0.19	0.23	0.26	0.30	0.34	0.38	0.43	0.47
% nonimmigrants		4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%
NonImmigrant Population (BoP)		2.30	2.58	2.79	3.19	4.05	3.53	4.33	5.15	6.03	6.95	7.96	9.03	10.14	11.29	12.49
(+) Nonimmigrant Visa Issued		10.4	9.7	9.0	8.7	4.0	2.8	4.4	5.7	7.0	8.3	9.1	10.0	10.9	11.7	12.6
% nonimmigrants		451%	375%	323%	273%	99%	79%	100%	110%	116%	119%	115%	111%	107%	104%	101%
% 2019 level					46%	32%	50%	65%	80%	95%	105%	115%	125%	135%	145%	
(-) Churn		(10.1)	(9.5)	(8.6)	(7.8)	(4.5)	(2.0)	(3.5)	(4.8)	(6.0)	(7.3)	(8.1)	(8.9)	(9.7)	(10.5)	(11.4)
% nonimmigrants		(439%)	(367%)	(308%)	(246%)	(112%)	(56%)	(81%)	(93%)	(100%)	(104%)	(101%)	(99%)	(96%)	(93%)	(91%)
% 2019 level					58%	25%	45%	61%	77%	93%	103%	114%	124%	135%	145%	
NonImmigrant Population (EoP)		2.58	2.79	3.19	4.05	3.53	4.33	5.15	6.03	6.95	7.96	9.03	10.14	11.29	12.49	13.74
% immigrant population		5.9%	6.3%	7.1%	9.0%	8.0%	9.6%	11.1%	12.7%	14.3%	16.0%	17.7%	19.4%	21.2%	22.9%	24.6%
% y/y			8.1%	14.3%	27.0%	(13.0%)	22.8%	19.0%	17.0%	15.3%	14.6%	13.3%	12.3%	11.4%	10.7%	10.0%
% 2019 levels					87%	107%	127%	149%	172%	197%	223%	250%	279%	308%	339%	
Total Foreign-Born Population		43.7	44.5	44.8	44.9	44.1	45.3	46.4	47.5	48.7	49.9	51.0	52.2	53.4	54.6	55.9
% y/y			1.8%	0.7%	0.2%	(1.8%)	2.6%	2.5%	2.4%	2.4%	2.5%	2.2%	2.3%	2.3%	2.4%	2.4%

Appendix 10: Demand Part 4

Demand

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

		Historical						Estimates (Annual)									
Calendar Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Year		2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Fiscal Quarter		Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	
Fiscal Date	Units	2016A	2017A	2018A	2019A	2020A	2021A	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E	
DUOL LPR SAM		4.0	4.1	4.0	4.0	3.9	3.9	4.0	4.0	4.0	4.1	4.1	4.1	4.1	4.1	4.0	
% servicable		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	
DUOL Natural SAM		6.0	6.1	6.5	6.5	6.5	6.6	6.7	6.8	6.9	6.9	7.0	7.1	7.2	7.2	7.3	
% servicable		30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	
DUOL Unauthorized Immigrant SAM		0.8	0.8	0.6	0.6	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.5	0.5	0.5	0.4	
% servicable		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	
DUOL Nonimmigrant SAM		0.9	0.9	1.1	1.5	1.1	1.6	1.9	2.3	2.6	3.0	3.4	3.8	4.3	4.8	5.2	
% servicable (excl. students)		50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	50%	
Total DUOL Immigrant SAM		11.6	11.9	12.3	12.5	12.1	12.7	13.1	13.6	14.0	14.5	15.0	15.5	16.0	16.5	17.0	
% y/y			2.3%	3.5%	1.9%	(3.7%)	5.2%	3.4%	3.4%	3.3%	3.5%	3.1%	3.2%	3.2%	3.3%	3.3%	
Total US Work Learner SAM			16.4	17.1	17.5	13.1	13.7	15.8	17.3	18.3	19.2	19.9	20.7	21.5	22.3	23.1	
% y/y				4.2%	2.5%	(24.7%)	4.1%	15.8%	9.0%	6.2%	4.4%	3.9%	3.9%	3.8%	3.7%	3.6%	
% 2019 levels						75%	78%	91%	99%	105%	110%	114%	118%	123%	127%	132%	
US MAUs (Using for Work)		1.3	1.4	1.5	1.7	2.2	2.0	2.2	2.5	2.8	3.1	3.3	3.6	3.9	4.2	4.6	
% SAM Penetration			8.4%	8.9%	9.8%	16.9%	14.4%	13.8%	14.6%	15.3%	16.1%	16.8%	17.6%	18.3%	19.1%	19.8%	
% y/y			9.6%	10.8%	13.0%	29.3%	(11.0%)	10.7%	14.9%	11.7%	9.6%	8.8%	8.5%	8.2%	7.9%	7.7%	

Appendix 11: Earnings History

Earning History (ERN)

NASDAQ: DUOL | FYE: December 31st | Unit in Millions except Per Share, Per Unit Data

	Correlation (Post Earnings)											Correlation (1W Prior to Earnings)										
	Revenue	Gross Profit	EBIT	EBITDA	Dil. EPS	MAU	Paid Subs	Subscription	Advertising	English Test	Other Rev	Revenue	Gross Profit	EBIT	EBITDA	Dil. EPS	MAU	Paid Subs	Subscription	Advertising	English Test	Other Rev
1D	(0.01)	0.23	(0.27)	(0.22)	(0.30)	0.59	(0.21)	(0.22)	0.35	0.10	0.29	0.89	0.68	0.98	0.93	0.99	(0.96)	(0.76)	(0.57)	0.53	0.85	(0.57)
1W	0.16	0.51	(0.23)	(0.32)	(0.21)	0.29	(0.05)	0.21	0.66	0.19	(0.33)	(0.73)	(0.51)	(0.84)	(0.93)	(0.81)	0.50	0.95	0.98	(0.39)	(0.80)	(0.04)
1M	0.32	0.65	(0.07)	(0.20)	(0.05)	0.09	(0.13)	0.19	0.78	0.33	(0.52)	0.04	0.40	(0.33)	(0.49)	(0.30)	0.19	0.24	0.56	0.55	0.01	(0.54)

=VADData(VA_Ticker,Time,Line Item,"Consensus/consensus.va_actuals/preQ",SD/CD,)

Time	Fiscal Date	Actuals											Consensus PreQ										
		Revenue	Gross Profit	EBIT	EBITDA	Dil. EPS	MAU	Paid Subs	Subscription	Advertising	English Test	Other Rev	Revenue	Gross Profit	EBIT	EBITDA	Dil. EPS	MAU	Paid Subs	Subscription	Advertising	English Test	Other Rev
8/4/22	Q2'22	88.4	64.5	4.5	4.2	0.1	49.5	3.3	65.2	11.2	8.0	3.9	85.7	62.5	(1.8)	(1.6)	(0.1)	46.3	3.1	64.5	11.1	7.5	2.9
5/12/22	Q1'22	81.2	59.7	4.3	3.9	0.1	49.2	2.9	58.0	11.7	8.1	3.4	77.6	56.5	(3.6)	(3.3)	(0.1)	45.9	2.7	58.0	9.5	6.1	2.4
3/3/22	Q4'21	73.0	53.1	(2.6)	0.3	(0.1)	42.4	2.5	51.1	11.1	8.1	2.7	68.9	49.6	(4.5)	(4.3)	(0.1)	42.0	2.4	50.9	8.4	5.6	4.1
11/10/21	Q3'21	63.6	45.5	(5.7)	(6.0)	(0.2)	41.7	2.2	46.0	9.0	6.7	1.8	60.5	43.5	(10.2)	(9.8)	(0.4)	40.5	2.1	46.2	8.3	5.0	1.3
8/11/21	Q2'21	58.8	42.7	3.4	3.7	0.2	37.9	1.9	43.5	9.1	4.8	1.4	-	-	-	-	-	-	-	-	-	-	-

Appendix 12: Employee Data Part 1

Employee Data

	Historical			Quarters														
Calendar Year	12/31/19	12/31/20	12/31/21	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	9/30/22
Year	2019	2020	2021	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022
Fiscal Year	2019	2020	2021	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022
Fiscal Quarter	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'
Fiscal Date	2019A	2020A	2021A	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22
Engineering				53%	50%	50%	50%	50%	50%	49.4%	48.6%	47.7%	46.9%	46.0%	45.1%	45%	45%	45%
Community				5%	6%	6%	6%	6%	6%							-	-	-
Research				3%	4%	4%	4%	4%	4%							8%	8%	8%
Research				8%	10%	10%	10%	10%	10%	12.0%	12.5%	11.8%	12.0%	11.0%	10.0%	9%	8%	8%
Employees (R&D Costs)				61%	60%	60%	60%	60%	60%	61%	61%	60%	59%	57%	55%	54%	53%	53%
Marketing				3%	6%	6%	6%	6%	6%	6.0%	6.3%	6.6%	6.9%	7.1%	7.4%	8%	8%	8%
Sales				3%	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Employees (S&M)				6%	6%	6%	6%	6%	6%	6.0%	6.3%	6.6%	6.9%	7.1%	7.4%	8.0%	8.0%	8.0%
People & Leadership				3%	4%	4%	4%	4%	4%	4.9%	5.9%	6.8%	7.7%	8.6%	9.6%	11%	11%	11%
Product Management				3%	-	-	-	-	-	1.4%	2.7%	4.1%	5.4%	6.8%	8.1%	10%	10%	10%
Design				17%	15%	15%	15%	15%	15%	13.3%	12.3%	12.0%	11.1%	10.9%	10.7%	10%	10%	10%
Operations				8%	15%	15%	15%	15%	15%	13.0%	11.8%	11.1%	10.0%	9.5%	9.0%	8%	8%	8%
Employees (G&A)				31%	34%	34%	34%	34%	34%	33%	33%	34%	34%	36%	37%	38%	39%	39%
Total %				98%	100%	100%	100%	100%	100%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Total Employees	350	500		-	-	-	-	-	-	-	-	400	400	400	500	500	-	-
Research Employees				17	23	23	24	27	28	43	48	48	55	55	53	53	53	53
(-) UX Designers				(1)	(1)	(1)	(1)	(3)	(3)	(3)	(3)	(3)	(4)	(4)	(4)	(4)	(4)	(4)
Research Employees (Adjusted)				16	22	22	23	24	25	40	45	45	51	51	49	49	49	49
Implied Total Employees				200	220	220	230	240	250	333	360	381	425	464	490	544	613	613
% q/q					10.0%	-	4.5%	4.3%	4.2%	33.3%	8.0%	5.9%	11.4%	9.1%	5.7%	11.1%	12.5%	-
% y/y								9%	14%	45%	50%	53%	28%	29%	28%	28%	32%	25%

Appendix 12: Employee Data Part 2

Employee Data

Calendar Year	Historical			Quarters														
	12/31/19	12/31/20	12/31/21	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	9/30/22
Year	2019	2020	2021	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022
Fiscal Year	2019	2020	2021	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022
Fiscal Quarter	Q4'	Q4'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'
Fiscal Date	2019A	2020A	2021A	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22
Implied Total Employees				200	220	220	230	240	250	333	360	381	425	464	490	544	613	613
% q/q					10.0%	-	4.5%	4.3%	4.2%	33.3%	8.0%	5.9%	11.4%	9.1%	5.7%	11.1%	12.5%	-
% y/y								9%	14%	45%	50%	53%	28%	29%	28%	28%	32%	25%
Research Employees (Adjusted)				16	22	22	23	24	25	40	45	45	51	51	49	49	49	49
Engineers				106	110	110	115	120	125	165	175	182	199	213	221	245	276	276
R&D Employees				122	132	132	138	144	150	205	220	227	250	264	270	294	325	325
S&M Employees				12	13	13	14	14	15	20	23	25	29	33	36	44	49	49
People & Leadership				6	9	9	9	10	10	16	21	26	33	40	47	57	67	67
Product Management				6	-	-	-	-	-	5	10	16	23	31	40	52	61	61
Design				34	33	33	35	36	38	44	44	46	47	51	53	54	61	61
Operations				16	33	33	35	36	38	43	42	42	43	44	44	44	49	49
G&A Employees				62	75	75	78	82	85	109	118	129	146	166	183	207	239	239
Total Employees				196	220	220	230	240	250	333	360	381	425	464	490	544	613	613

Appendix 13: User Base Webscraped

	Historicals							
	12/31/15	12/31/16	12/31/17	12/31/18	12/31/19	12/31/20	12/31/21	12/31/22
	2015	2016	2017	2018	2019	2020	2021	2022
	2015	2016	2017	2018	2019	2020	2021	2022
	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'	Q4'
Cumulative	2015A	2016A	2017A	2018A	2019A	2020A	2021A	2022A
English Learners					74	83	96	124
% q/q								
% y/y						12%	15%	29%
Total Learners					179	217	240	297
% q/q								
% y/y						21%	11%	23%
% English Share					41.6%	38.4%	39.8%	41.7%
bps q/q								
bps y/y						(318)	137	196
Actual MAUs					23	27	37	40
% q/q								
% y/y						17%	34%	10%
Learner to MAU Ratio					7.7x	7.9x	6.6x	7.3x
Incremental								
Incremental English Learners						9	12	28
% q/q								
% y/y							41%	128%
Incremental Learners						38	24	56
% q/q								
% y/y							(37%)	138%
% Incremental English Share						23.3%	52.3%	50.1%
bps q/q								
bps y/y							2,907	(225)

Appendix 14: Wage Data

Wage Data

Glassdoor Data as of 9/29/22

Operations Salary		
Job Title	Salary	Number Value
Business C	\$125,534/yr	\$125,534 125,534
Senior Op	\$121,021/yr	\$121,021 121,021
Senior Prc	\$136,228/yr	\$136,228 136,228
Content C	\$66,293/yr	\$66,293 66,293
Operation	\$77,163/yr	\$77,163 77,163
Vice Presi	\$194,451/yr	\$194,451 194,451
Average		120,115

Research		
Duolingo Salaries		
Job Title	Salary	
Learning S	\$102,995/yr	\$102,995 102,995
Data Scier	\$123,447/yr	\$123,447 123,447
Data Scier	\$128,086/yr	\$128,086 128,086
Senior Dai	\$145,913/yr	\$145,913 145,913
Learning S	\$103,805/yr	\$103,805 103,805
Principal S	\$150,977/yr	\$150,977 150,977
Staff Data	\$160,337/yr	\$160,337 160,337
		135,428

SWE		
Duolingo Salaries		
Job Title	Salary	
Software I	\$132,810/yr	\$132,810 132,810
Senior Sof	\$142,739/yr	\$142,739 142,739
Software I	\$130,071/yr	\$130,071 130,071
Software I	\$133,150/yr	\$133,150 133,150
Staff Softw	\$154,684/yr	\$154,684 154,684
SWE salari	\$173,162/yr	\$173,162 173,162
Software I	\$103,910/yr	\$103,910 103,910
Developm	\$127,326/yr	\$127,326 127,326
Engineeri	\$203,040/yr	\$203,040 203,040
IOS Engin	\$135,510/yr	\$135,510 135,510
Software I	\$124,979/yr	\$124,979 124,979
Software I	\$132,804/yr	\$132,804 132,804
		141,182

Product Management		
Job Title	Salary	
Product M	\$137,625/yr	\$137,625 137,625
Senior Prc	\$158,122/yr	\$158,122 158,122
Program M	\$123,516/yr	\$123,516 123,516
Associate	\$101,478/yr	\$101,478 101,478
Project M	\$109,090/yr	\$109,090 109,090
Senior Prc	\$131,933/yr	\$131,933 131,933
Revenue I	\$132,405/yr	\$132,405 132,405
		127,738

People & Leadership		
Job Title	Salary	
Recruiter	\$76,195/yr	\$76,195 76,195
Senior Tec	\$102,324/yr	\$102,324 102,324
Human Re	\$110,796/yr	\$110,796 110,796
University	\$77,010/yr	\$77,010 77,010
Human Re	\$70,851/yr	\$70,851 70,851
Talent Acc	\$93,336/yr	\$93,336 93,336
		88,419

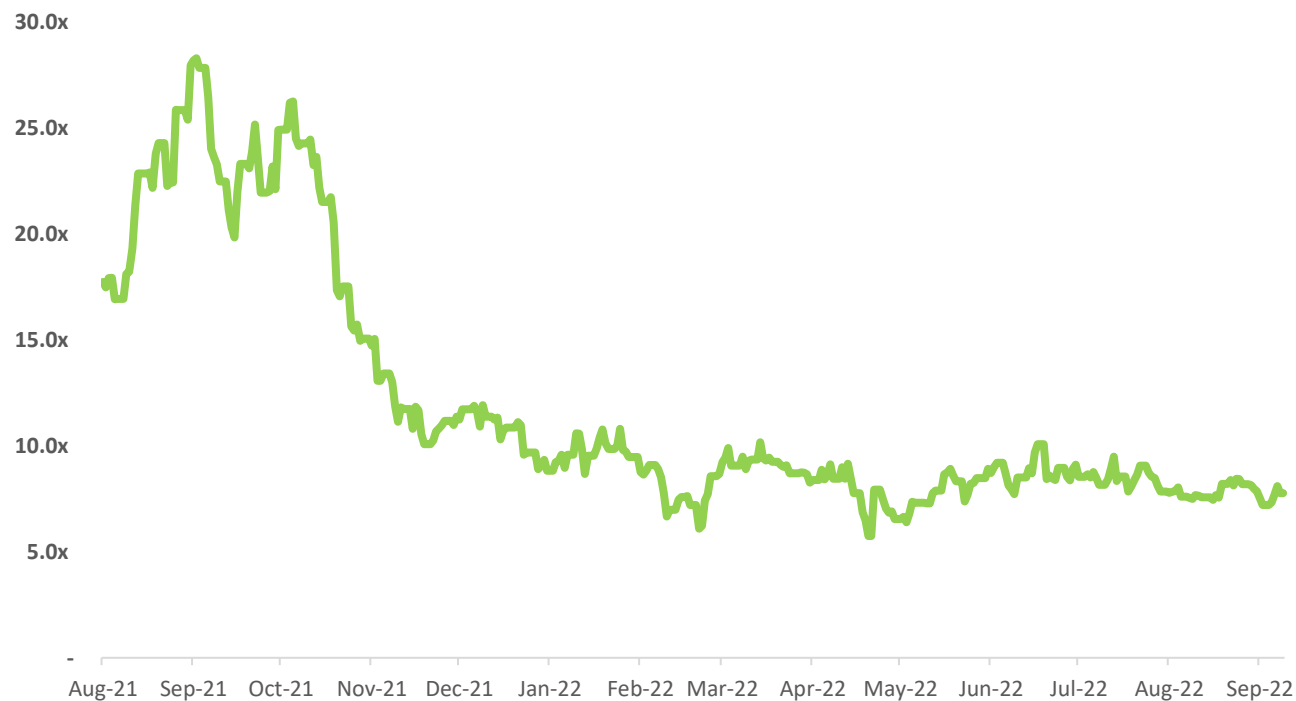
Business		
Job Title	Salary	
Chief Tec	\$209,422/yr	\$209,422 209,422
Senior HTI	\$136,386/yr	\$136,386 136,386
Senior Ma	\$150,960/yr	\$150,960 150,960
Director s	\$170,961/yr	\$170,961 170,961
Chief Mar	\$184,989/yr	\$184,989 184,989
Director, E	\$196,497/yr	\$196,497 196,497
		174,869

UX Design		
Job Title	Salary	
Product D	\$122,265/yr	\$122,265 122,265
Senior Prc	\$138,662/yr	\$138,662 138,662
Designer	\$102,370/yr	\$102,370 102,370
Lead Desig	\$112,173/yr	\$112,173 112,173
Content V	\$60,647/yr	\$60,647 60,647
Social Con	\$59,729/yr	\$59,729 59,729
Education	\$73,985/yr	\$73,985 73,985
Software	\$125,779/yr	\$125,779 125,779
Producer	\$97,273/yr	\$97,273 97,273
Content C	\$59,856/yr	\$59,856 59,856
		95,274

Appendix 15: Historical Multiple

Year	Month	NTM Multiple	
2021	Jun	-	
2021	Jul	-	
2021	Aug	5.6x	Aug-21
2021	Sep	24.5x	Sep-21
2021	Oct	23.3x	Oct-21
2021	Nov	17.2x	Nov-21
2021	Dec	11.2x	Dec-21
2022	Jan	10.0x	Jan-22
2022	Feb	9.6x	Feb-22
2022	Mar	8.1x	Mar-22
2022	Apr	8.9x	Apr-22
2022	May	7.3x	May-22
2022	Jun	8.4x	Jun-22
2022	Jul	8.8x	Jul-22
2022	Aug	8.2x	Aug-22
2022	Sep	7.8x	Sep-22
Average		11.4x	
Median		8.8x	
75th Percentile		10.9x	
25th Percentile		8.1x	

DUOL NTM Multiple Since IPO



Appendix 16: Inflation Data

Inflation Data

2017	2.1		3/31/18	6/30/18	9/30/18	12/30/18	3/30/19	6/30/19	9/30/19	12/30/19	3/30/20	6/30/20	9/30/20	12/30/20	3/30/21	6/30/21	9/30/21	12/30/21	3/30/22	6/30/22	9/30/22
Qtr3	2.1		2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022
Qtr4	2.1		2018	2018	2018	2018	2019	2019	2019	2019	2020	2020	2020	2020	2021	2021	2021	2021	2022	2022	2022
Qtr1	2.2		Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'	Q4'	Q1'	Q2'	Q3'
Qtr2	2.5		Q1'18	Q2'18	Q3'18	Q4'18	Q1'19	Q2'19	Q3'19	Q4'19	Q1'20	Q2'20	Q3'20	Q4'20	Q1'21	Q2'21	Q3'21	Q4'21	Q1'22	Q2'22	Q3'22
Qtr3	2.5	CPI	2.2	2.5	2.5	2.4	2.4	2.4	2.6	2.7	2.7	2.1	2.2	1.8	1.6	2.5	2.4	3.2	4.3	5.0	5.8
Qtr4	2.4	% form	2.2%	2.5%	2.5%	2.4%	2.4%	2.4%	2.6%	2.7%	2.7%	2.1%	2.2%	1.8%	1.6%	2.5%	2.4%	3.2%	4.3%	5.0%	5.8%
Qtr1	2.4																				
Qtr2	2.4																				
Qtr3	2.6																				
Qtr4	2.7																				
Qtr1	2.7																				
Qtr2	2.1																				
Qtr3	2.2																				
Qtr4	1.8																				
Qtr1	1.6																				
Qtr2	2.5																				
Qtr3	2.4																				
Qtr4	3.2																				
Qtr1	4.3																				
Qtr2	5.0																				
Qtr3	5.8																				

Appendix 17: Multiple Valuation

	Multiple Valuation				
	2022E	2023E	2024E	2025E	2026E
NTM Revenue (BIG)	\$496	\$639	\$793	\$939	\$1,082
% y/y	37%	29%	24%	18%	15%
(x) NTM Multiple	9.1x	8.1x	7.2x	6.0x	5.5x
Implied EV	4,517	5,152	5,720	5,633	5,952
(+) Cash	625	758	921	1,113	1,332
(-) Debt	26	26	26	26	26
Implied Market Cap	5,117	5,884	6,615	6,720	7,258
(/) FDSO	40	41	42	43	44
DUOL PT	\$129	\$144	\$157	\$155	\$163
% Δ	35%	51%	65%	63%	72%
% IRR	232%	39%	25%	16%	14%